

PR200 OPERATOR TRAINING HARDWARE INTRODUCTION

ROHDE & SCHWARZ

Make ideas real





PR200 OVERVIEW

PR200 OVERVIEW

- ▶ Detect, analyze and locate RF signals from **8 kHz to 8 GHz**; extendable up to **20 GHz** with the HE400DC & from **18 GHz to 33 GHz** with the HE800-DC30.
- ▶ Extensive preselection filtering and automatic overload protection
- ▶ High-speed panorama scanning with **up to 60 GHz/s** over the entire frequency range
- ▶ Simultaneous measurements in frequency and time domain with time-gated spectrum measurement
- ▶ Optimized for demanding field operation with minimal size, weight and power consumption
- ▶ Innovative application-based user interface for convenient, simple and intuitive operation



PR200 OVERVIEW

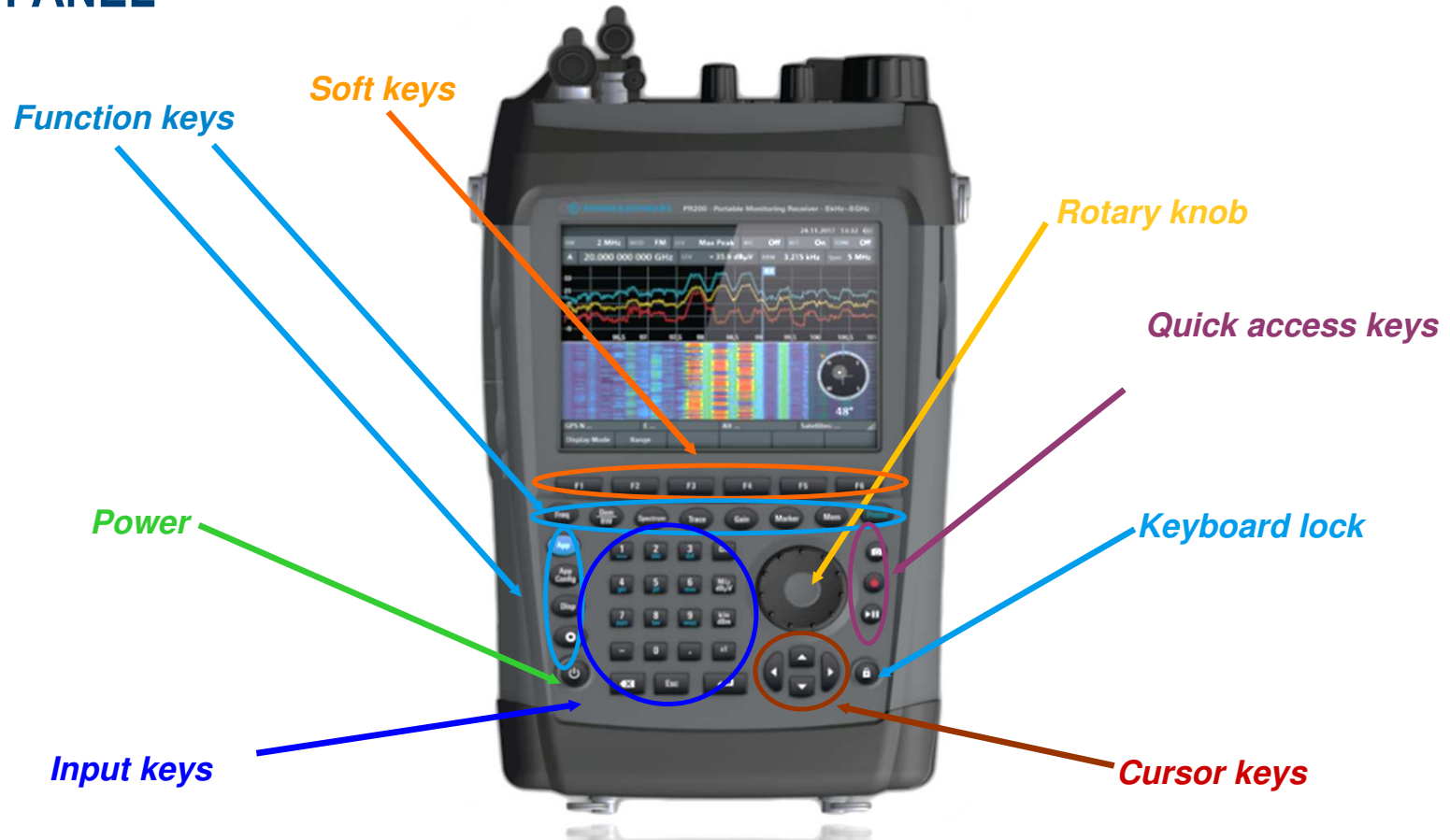
► PR200 basic Specifications

Technical Data

Frequency Range	8 kHz to 8 GHz
Demodulation modes	AM, FM, PM, pulse, I/Q, LSB, USB, CW, ISB
Bandwidth	10/30/50/100/150/200/300/600 Hz, 1/1.5/2.1/2.4/2.7/3.1/4/4.8/6/8.333/9/12/1 5/25/30/50/75/120/150/250/300/500/800 kHz, 1/1.25/1.5/2/5/8/10/12.5/15/20/40MHz
CISPR bandwidth	200 Hz, 9 kHz, 120 kHz, 1 MHz
Operating temperature	−10 °C to +55 °C
Dimensions & Weight	192 mm × 320 mm × 62 mm, 3.5 kg



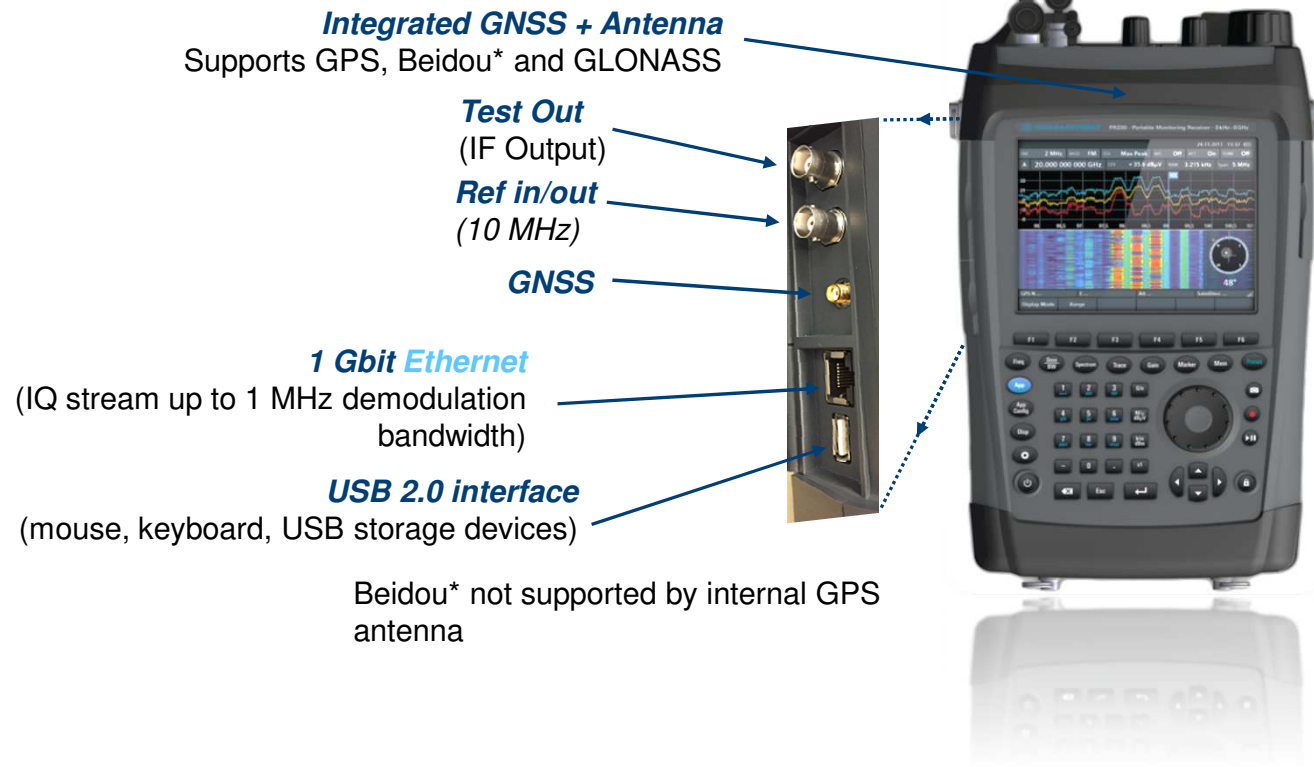
FRONT PANEL



EXTERNAL CONNECTORS AND PORTS - TOP



EXTERNAL CONNECTORS AND PORTS – TOP LEFT



EXTERNAL CONNECTORS AND PORTS - RIGHT





PR200 BASICS

BATTERIES AND CHARGING

- ▶ The PR200 can be operated from mains (AC) or from an internal battery pack.
- ▶ Average run time is approximately 3 to 3.5 hours depending on operating modes.
- ▶ Time to charge is ~7 hours while unit is running, ~3 hours with unit off
- ▶ External battery charger and extra batteries are available.



BATTERIES AND CHARGING

■ Charge via device



Note: use only supplied power adaptor for charging.

■ Charge via external battery charger



Battery Icon



LED indicator

Power button LED indicator

Charging (blinking)
Fully charged (continuous)



Standby mode,
no battery connected



Power is ON

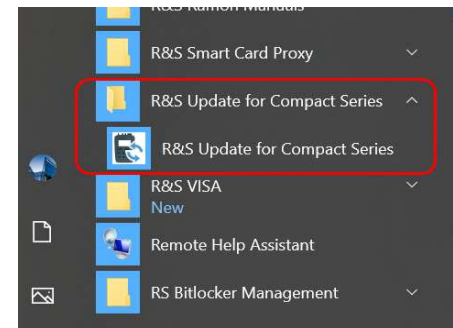
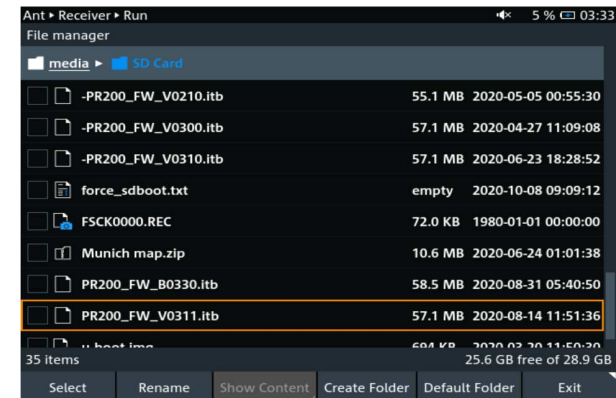


Error in the device
or battery error charging



UPDATING FIRMWARE

- Firmware update methods:
 - 1) SD card (root directory)
 - 2) USB memory stick (root directory)
 - 3) remote update via software on PC i.e. R&S Update for Compact Series, InstrumentView.
- If using USB stick, ensure no other firmware file in SD card (recommended to remove SD card).
- When using SD card or USB memory stick, press and hold the power button for 5 seconds to boot up the device.
- Battery level has to be > 30% else power adaptor is needed



FACTORY RESET

To reset the R&S PR200 back to factory settings:

Switch on the R&S PR200.

Press and hold the power button (1) for 4 seconds.

R&S PR200 automatically performs the factory reset procedure.

During a factory reset:

All apps are removed from the softkeys and the "Receiver" app is assigned to the second softkey (F2).

NOTE: F1 is the "Edit" key.

All settings are returned to the default values.

All list entries are deleted.

(i.e. "Memory List", "Positions List" and "Suppress List").

Note: All lists can be exported as backup using the "Manage" function.



STEALTH MODE

To avoid unwanted attention when performing measurements, the screen can be switched off and the controls can be locked.

To switch Stealth mode on,

Press the Lock button (1) which displays an option dialog and select Stealth mode.

The display screen switches off and the controls lock upon confirming the selection.

To turn off Stealth mode,

Press the Lock button (1) three times.



HE400 Overview





ANTENNA OVERVIEW – HE400

HE400 OVERVIEW

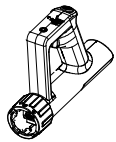
- ▶ The R&S HE400 handheld directional antenna is primarily intended for locating transmitters and interference sources by pointing the antenna towards the direction of maximum signal strength.
- ▶ It consists of an antenna handle, a receiver-customized cable set, and different antenna modules, which can be easily attached by means of turning the locking ring.
- ▶ A built-in low-noise broadband amplifier (LNA) can be activated (active mode = LNA ON) in order to increase the system sensitivity.
- ▶ In passive mode (LNA OFF), the amplifier is bypassed and may also be used in the vicinity of strong transmitters



HE400 PRODUCT FAMILY OVERVIEW

Antenna Handles:

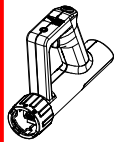
HE400 Handheld Dir. Antenna
4104.6000.02
8.3kHz - 8GHz
Receivers: PR100, PR200, FSH4, FSH8, FPH



Antenna Modules:
HE400HF
HE400VHF
HE400CEL
HE400UWB
HE400LP
HE400SCB
HE400SHF

Cables:
HE400-K Cable Set PR100 HE400
HE400-K Cable Set PR200 HE400

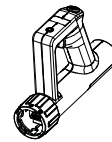
HE400MW Microwave Handheld Dir. Ant.
4104.6000.03
20MHz - 20GHz
Receivers: FSH20, Callisto Microwave, PR100, PR200



Antenna Modules:
HE400VHF
HE400CEL
HE400UWB
HE400LP
HE400SCB
HE400SHF

Cables:
HE400-K Cable Set PR100 HE400
HE400-K Cable Set PR200 HE400

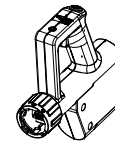
HE400BC Basic Handheld Dir. Ant.
4104.6000.04
8.3kHz - 20GHz
Receivers: OEM



Antenna Modules:
HE400HF
HE400VHF
HE400UWB
HE400LP
HE400SHF

Cables:
HE400-KB Cable for HE400BC

HE400DC Handheld Dir. Ant. w. Down Converter
4104.6000.05
20MHz - 20GHz
Receivers: PR200

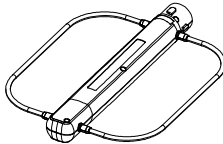


Antenna Modules:
HE400VHF
HE400CEL
HE400UWB
HE400LP
HE400SCB
HE400SHF

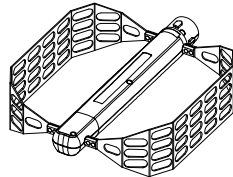
Cables:
HE400-K Cable Set PR200 HE400

Antenna Modules:

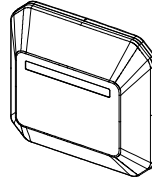
HE400HF HF Module
4104.8002.02
8.3kHz - 30MHz



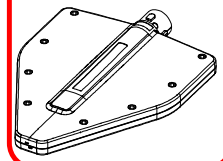
HE400VHF VHF Module
4104.8202.02
20MHz - 200MHz



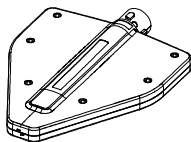
HE400CEL Cellular Module
4104.7306.02
700MHz - 2.5GHz



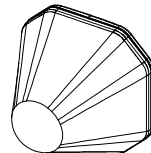
HE400UWB UWB Module
4104.6900.02
30MHz - 6GHz



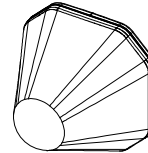
HE400LP Log.Per. Module
4104.8402.02
450MHz - 8GHz



HE400SCB S/C Band Module
4104.7606.02
1.7GHz - 6GHz



HE400SHF SHF Module
4104.8602.02
5GHz - 20GHz



HE400DC – HANDHELD DIRECTIONAL ANTENNA WITH DOWNCONVERTER



- ▶ Extremely broadband due to several antenna modules
- ▶ Wide dynamic range due to switchable active and passive mode (LNA ON/OFF)
- ▶ Vertical and horizontal polarization by turning of antenna module
- ▶ Automatic antenna module and polarization recognition
- ▶ Handy size, light weight
- ▶ Can be mounted on a tripod (1/4" tripod thread)
- ▶ Trigger button for external triggering of the receiver
- ▶ GPS/GLONASS receiver and electronic compass included

HE400DC – HANDHELD DIRECTIONAL ANTENNA WITH DOWNCONVERTER

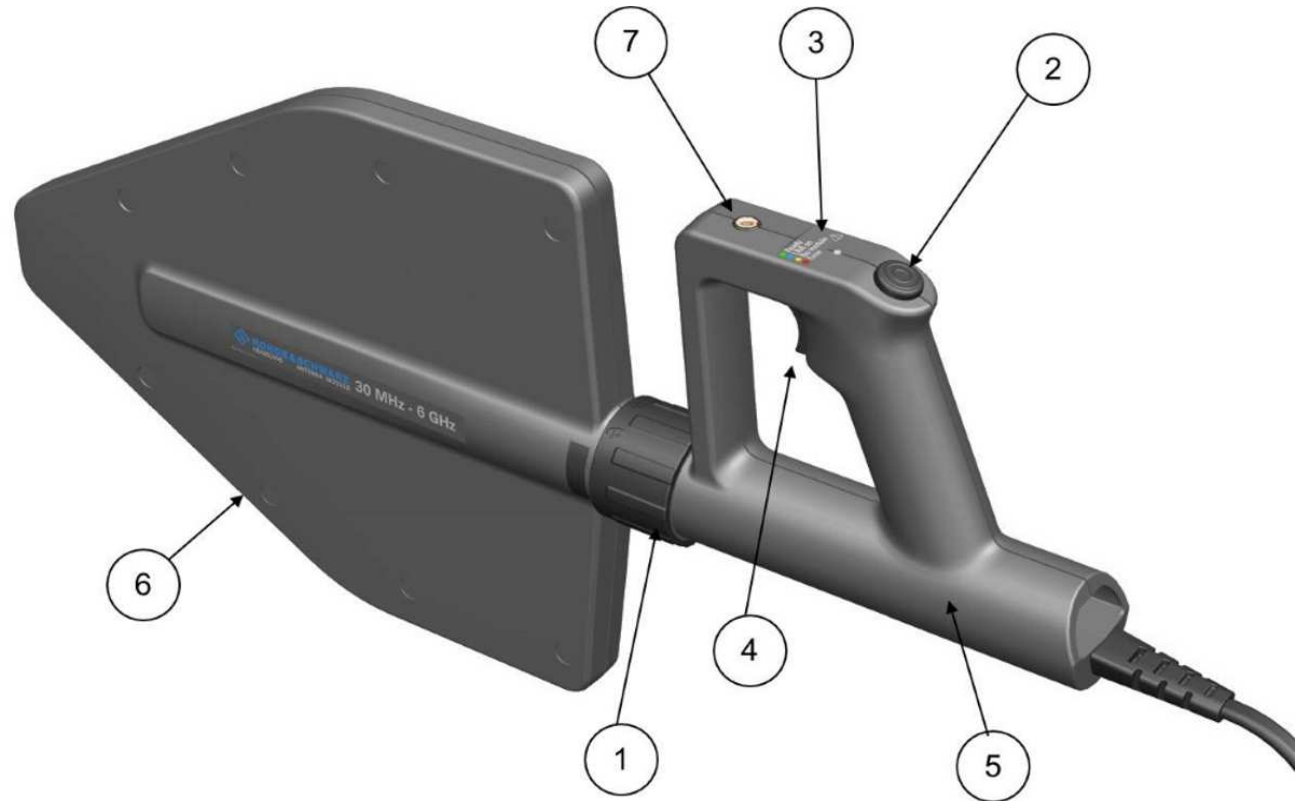


Technical data

RF input frequency range	20 MHz to 20 GHz*
Conversion gain (LNA active)	Typ. 13 dB
Output VSWR (including RF cable)	< 3.0, typ. 2.0
Polarization	Adjustable, linear vertical or horizontal
Extra features	GNSS electronic compass

HANDHELD DIRECTIONAL ANTENNA - OVERVIEW

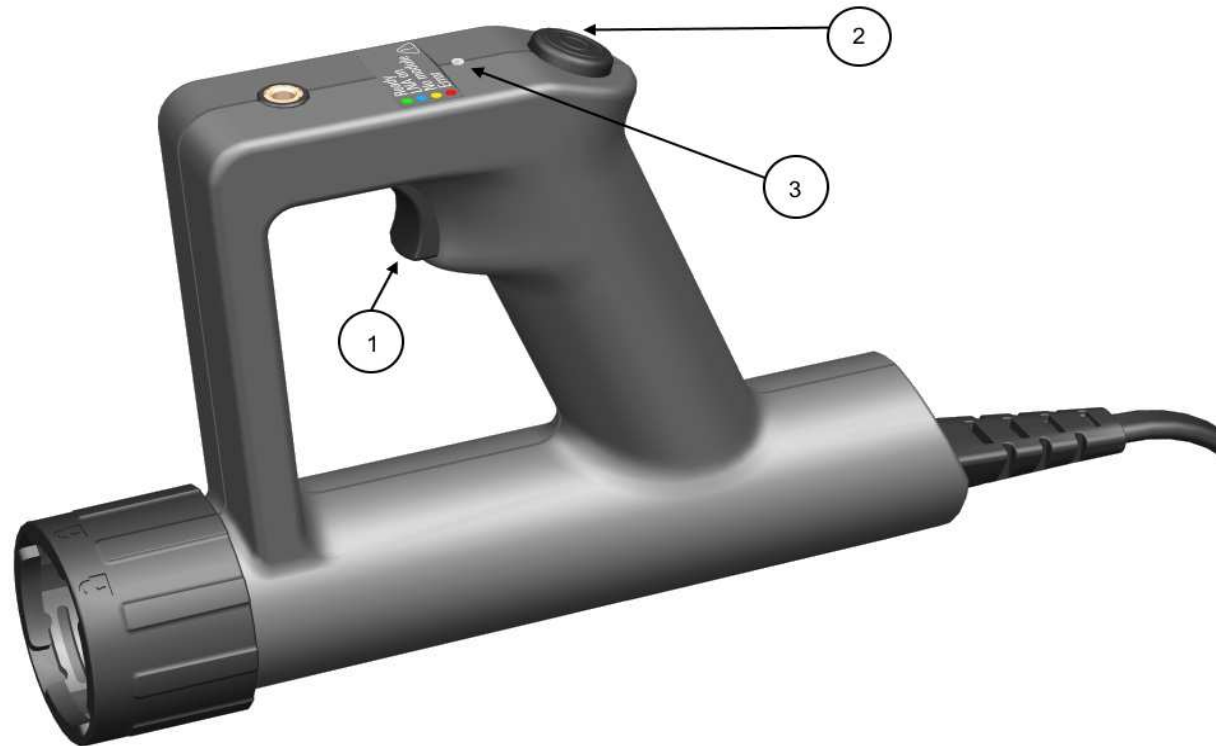
- ▶ 1 = Locking ring
- ▶ 2 = Toggle button
- ▶ 3 = Status LED
- ▶ 4 = Trigger button
- ▶ 5 = Antenna handle
- ▶ 6 = Antenna module
- ▶ 7 = Adapter thread



HANDHELD DIRECTIONAL ANTENNA - BUTTONS & LED

1=Trigger button

- ▶ Short press:
 - Generate trigger signal to the receiver
 - Stop compass calibration
- ▶ Used in triangulation mode (Map function) to save the current compass and position with a single press.
- ▶ Used in HSCAN mode to start/pause with a single press or save the measurement with a double presses.



HANDHELD DIRECTIONAL ANTENNA - BUTTONS & LED

2=Toggle button

3=Status LED



- ▶ **Green Steady** Valid antenna module - passive mode (LNA OFF).
- ▶ **Blue Steady** Valid antenna module - active mode (LNA ON).
- ▶ **Yellow Steady** No valid antenna module detected.
- ▶ **Red Steady** Test point failure
- ▶ **Magenta Flashing:**
 - Compass calibration routine in progress.
 - Firmware update in progress



HANDHELD DIRECTIONAL ANTENNA - BUTTONS & LED

With the toggle button, you can switch the LNA control (ON/OFF),

1. LNA OFF (passive mode): The low-noise amplifier is bypassed and has no supply voltage.
2. LNA ON (active mode): The low-noise amplifier is enabled and the status LED lights up blue,

LNA ON is only recommended:

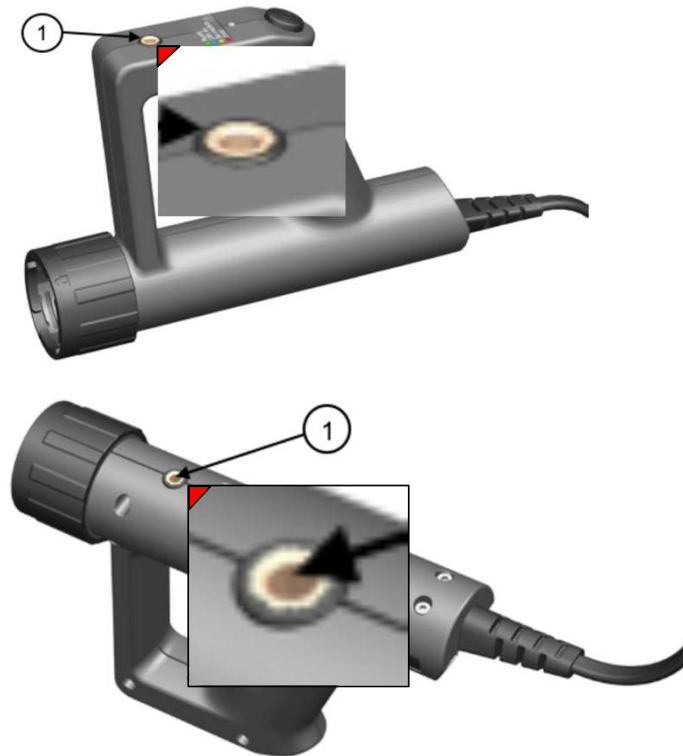
If there are no strong transmitters in close vicinity

If the signal is not sufficient to detect with LNA OFF

3. The active mode (LNA ON) can only be activated if a valid antenna module is connected to the antenna handle.
4. The antenna automatically switches to the bypass mode if the antenna module is removed from the antenna handle.

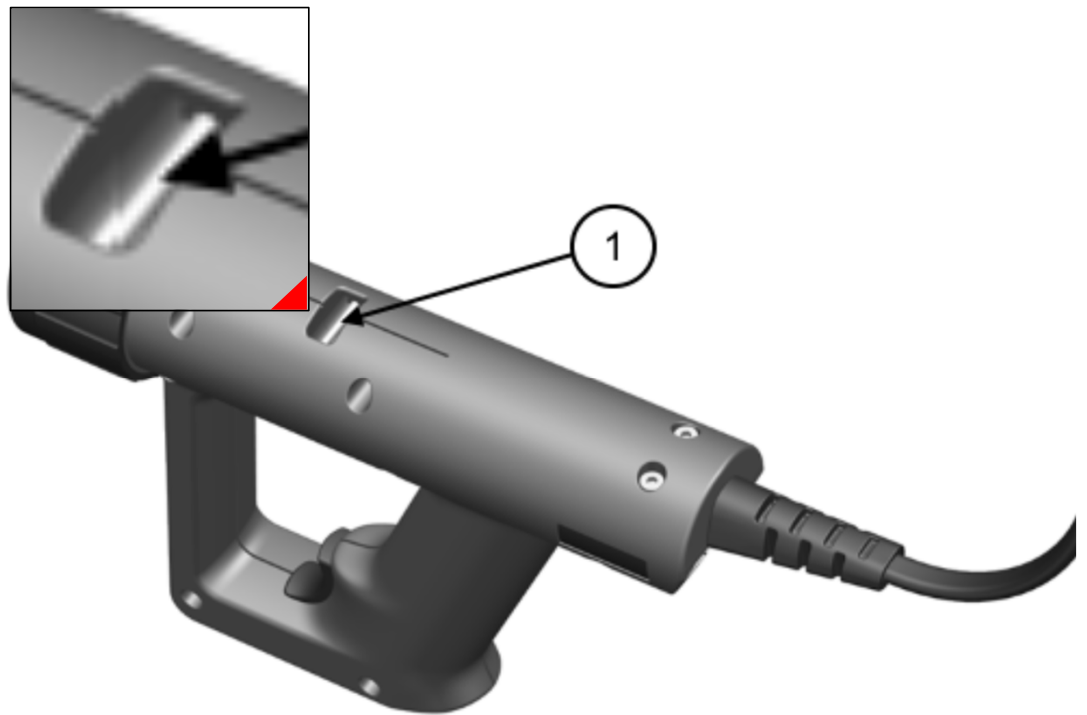
HANDHELD DIRECTIONAL ANTENNA - MOUNTING THREAD

- ▶ The R&S HE400 has a 1/4" adapter thread on top of the antenna handle
- ▶ Suitable for attaching a smartphone or tablet holder
- ▶ The R&S HE400 has a 1/4" tripod thread in the lower part of the antenna handle
- ▶ Suitable for attachment to a commercial tripod.



HANDHELD DIRECTIONAL ANTENNA - STRAP PIN

- ▶ The R&S HE400 has an attachment pin in the lower part of the antenna handle
- ▶ Suitable for attaching a strap.
- ▶ Useful for carrying the antenna when you need both hands free

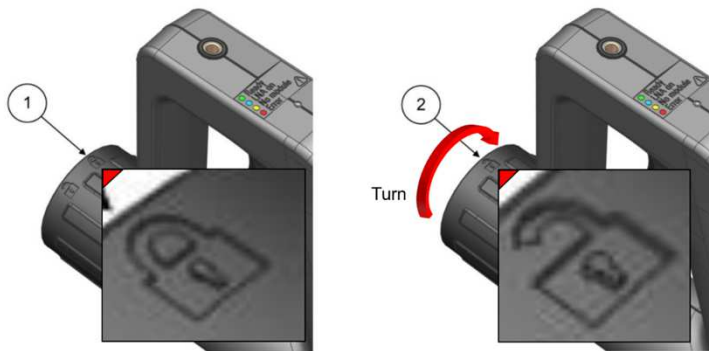




HARDWARE SETUP

PREPARING THE R&S HE400/800 ANTENNA

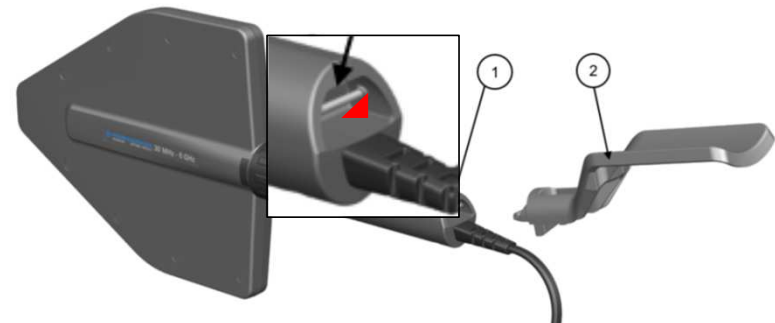
- Make sure that the locking ring on the antenna handle is in unlocked by turning the locking ring.



- Insert the desired antenna module.

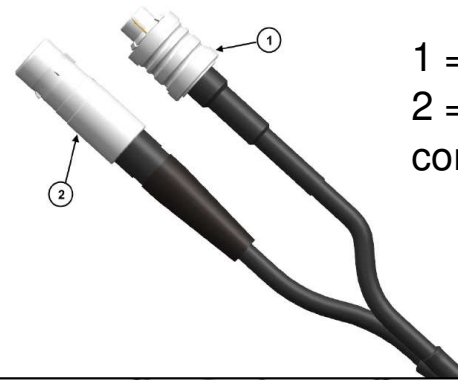


- Turn the locking ring back to the locked position”.
- Insert the arm rest into the handle and push it down to lock it into place .

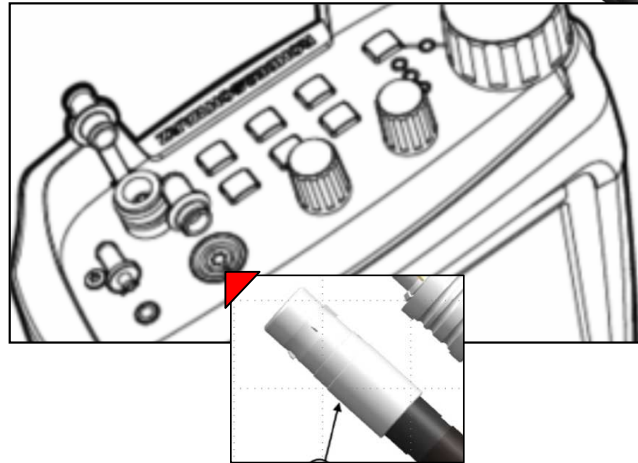
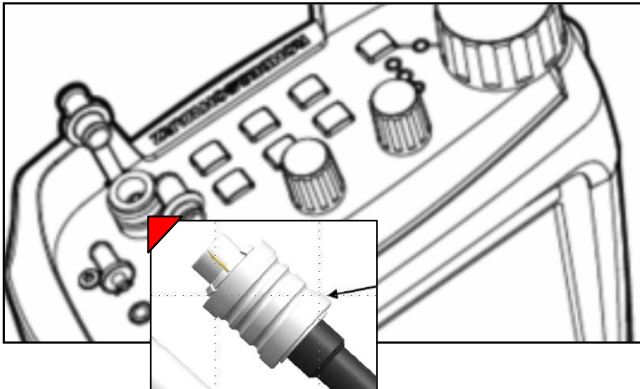


PREPARING THE R&S HE400/800 ANTENNA- CONNECTING TO PR200

- ▶ Connect the RF connector to connector ANT
- ▶ Connect the supply and control connector to connector AUX1

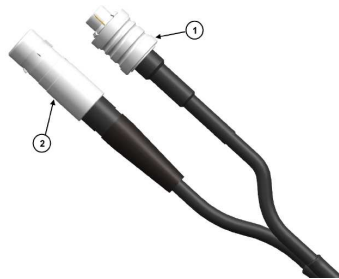


1 = RF connector
2 = Supply and control
connector

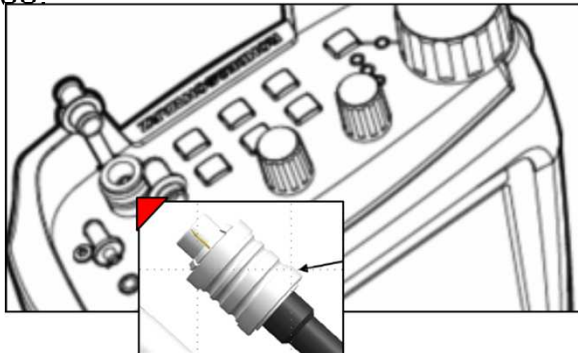


PREPARING THE R&S HE400/800 ANTENNA- CONNECTING TO PR200

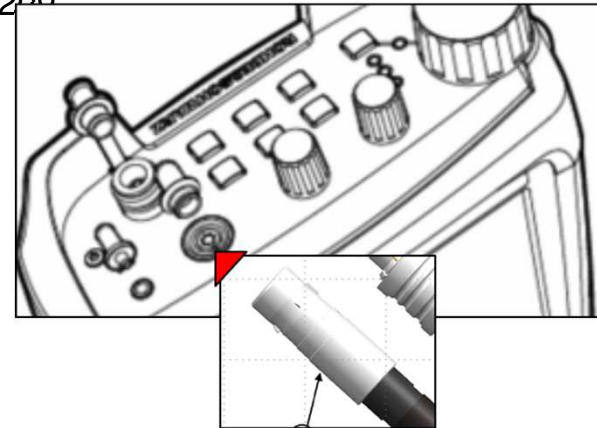
1. Ensure that R&S PR200 is powered off.
2. Connectors of R&S HE400 consists of a SnapN type RF connector and an ODU connector.



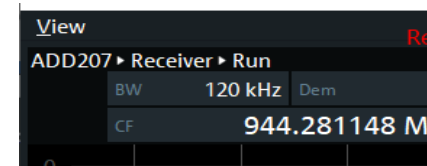
3. Push the SnapN RF connector into ANT port of R&S PR200.



4. Push the ODU connector into AUX1 port of R&S PR200.



5. Power on R&S PR200.
6. In the top left corner of the receiver mode display, it shows the antenna connected to it. (e.g. ADD207 connected)



PREPARING THE PR200 - INSERTING THE BATTERY

The battery is in a stand-by state for shipment. To revive the battery, insert it into the R&S PR200 and connect the power adaptor

1. Slide the battery cover downwards to open it.
2. Insert the battery (electrical contacts first) into the compartment.
The grooves on the battery make it impossible to insert incorrectly.
3. Fold battery cover back into the closed position.
4. Push battery cover upwards to lock it.
5. Connect the power adapter to the connector on the left side of the R&S PR200 to charge the battery.
6. Charge the battery to maximum.
Charging can take a few hours to complete.



PREPARING THE PR200 - TURNING ON AND BASIC SYSTEM CHECKS

1. Press the power button to turn on the RP200. (do note the color of the power button) wait till the button shines GREEN to indicate the device is on

Power button LED indicator

Charging (blinking)
Fully charged (continuous)



Power is ON



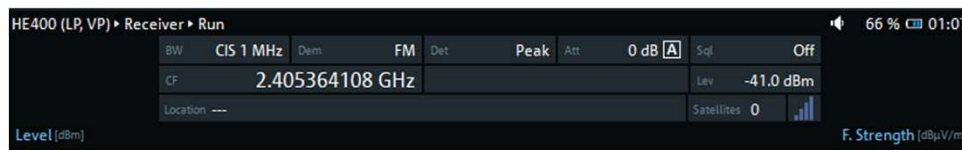
Standby mode,
no battery connected



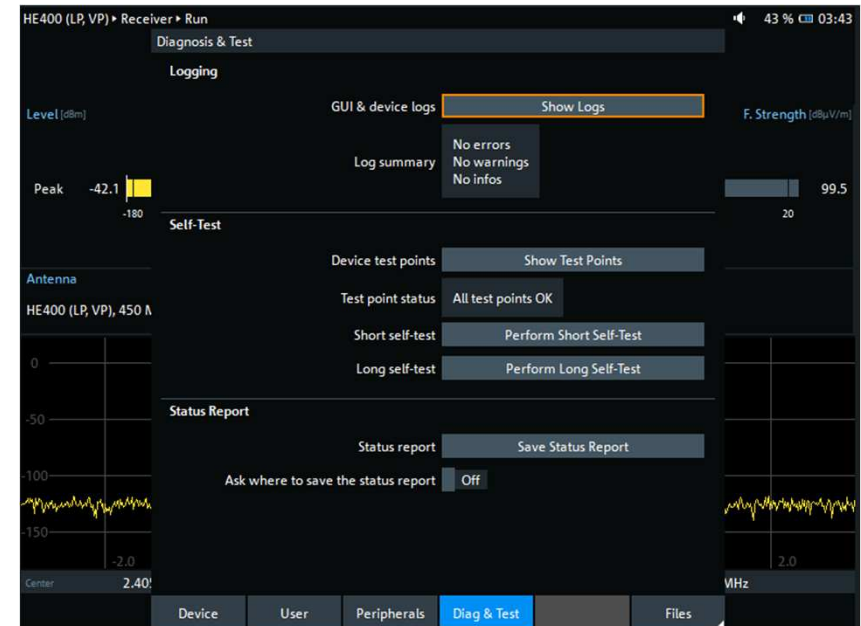
Error in the device
or battery error charging



2. After the PR200 has fully booted, check that the receiver recognize the HE400/HE800 antenna

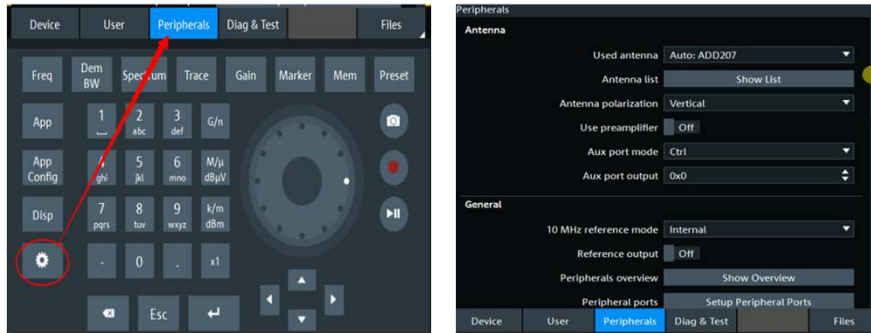


3. Under settings, , “Diag & Test” check that there are no additional errors spotted. Ensure also that all test points are OK as well.

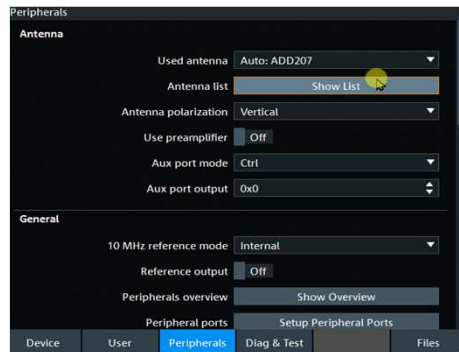


SETTING TO USE ANTENNA COMPASS AND GPS

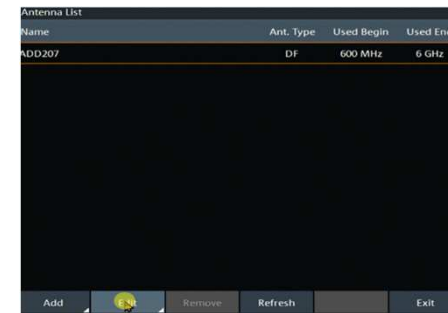
1. Press [Device Config] and then “Peripherals”. The “Peripherals” page appears in the display.



2. In the “Antenna” section, select “Show List” and press the center of rotary control.



3. Select the desired antenna and press the “Edit” button.

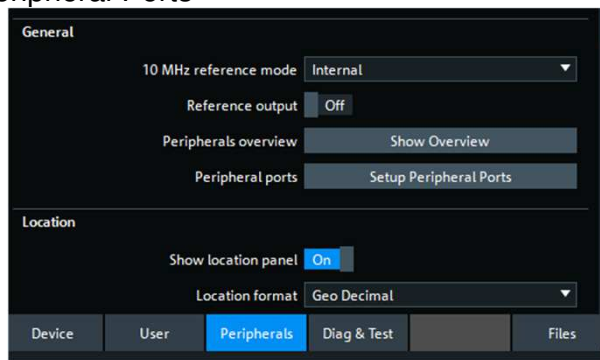


4. Scroll to “GNSS & Compass” section. Enable both “Antenna GNSS” and “Antenna Compass”.

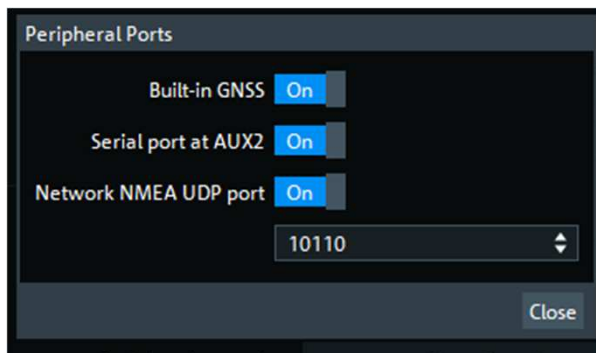


SETTING TO USE ANTENNA COMPASS AND GPS

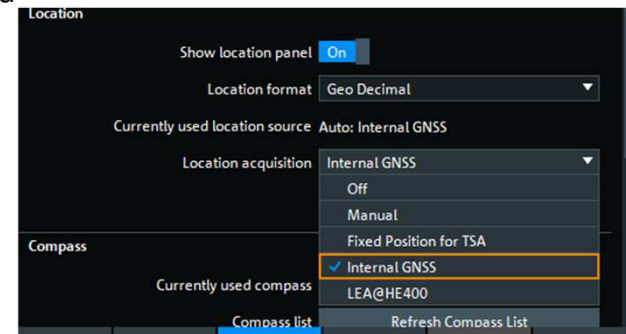
1. To use the internal GNSS built into the PR200, first ensure it is enabled. Go to “setting”, “Peripherals”. Under General, select “Setup Peripheral Ports”



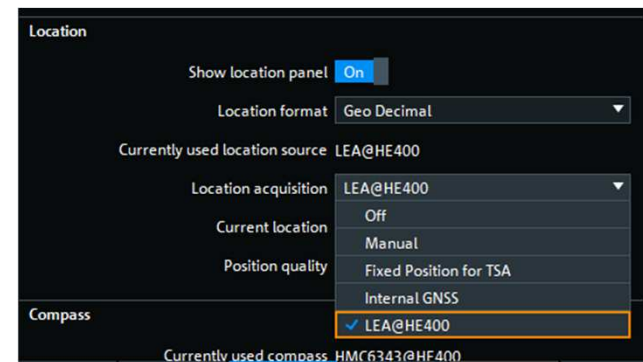
2. Make sure that the Built-in GNSS is turned on



3. Scroll to “Location” section. Under “Location acquisition”, ensure that the Internal GNSS is selected in the dropdown menu

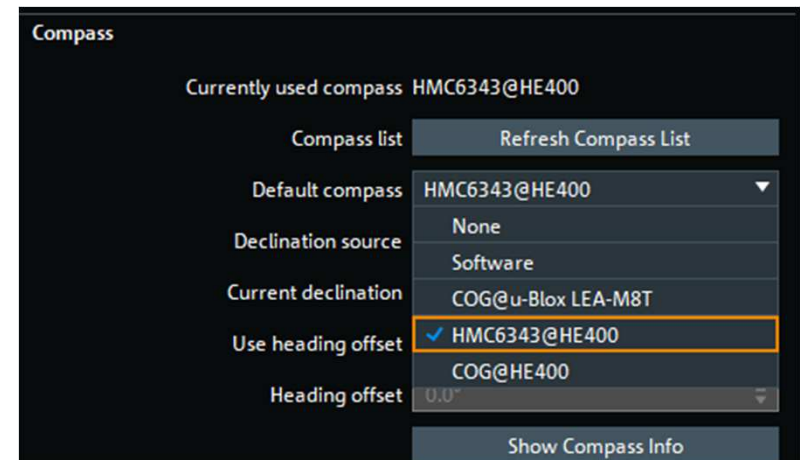
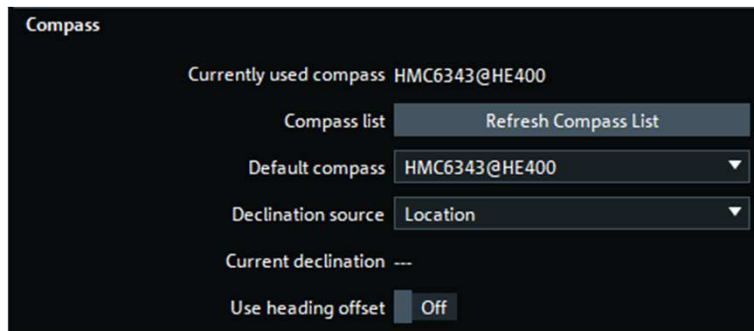


4. To use the GNSS Antenna built into the HE400, select the “LEA@HE400” instead



SETTING TO USE ANTENNA COMPASS AND GPS

1. To use the compass built into the PR200, under “setting”, “Peripherals”. Under Compass, make sure that “HMC6343@HE400” is selected for the currently used compass.
2. If not selected, ensure that the right GPS antenna is selected before.
3. Select the “HMC6343@HE400” from the dropdown under “Default compass”



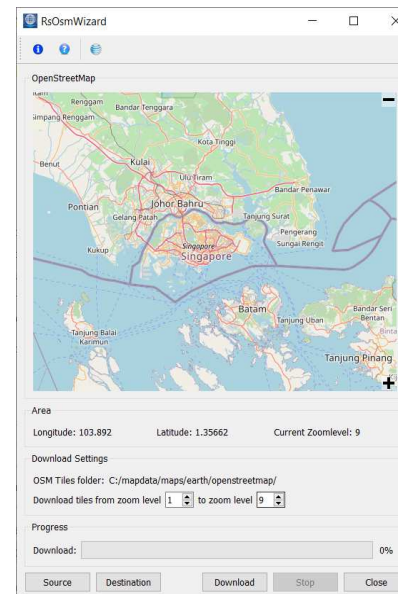
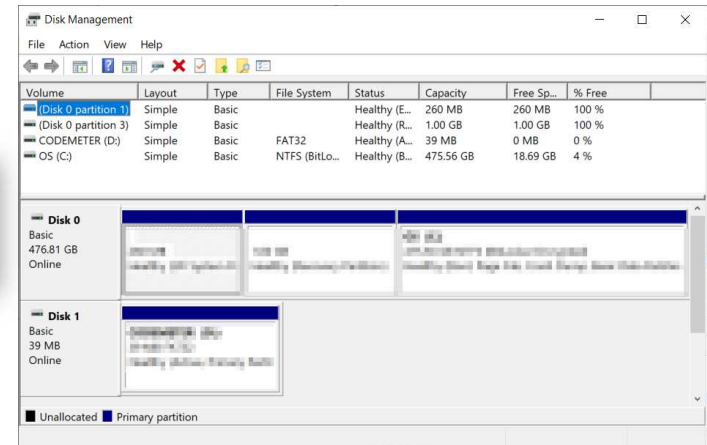
PREPARING SD CARD

Before using a new SD card or USB storage device, it is strongly recommended to format it as FAT32 or exFAT. To perform the formatting run diskmgmt.msc in Windows OS.

Map view function is available with CS-MAP option and a world map is pre-installed with 2 zoom levels.

Digital map data of the area of interest has to be downloaded and copied to the SD card.

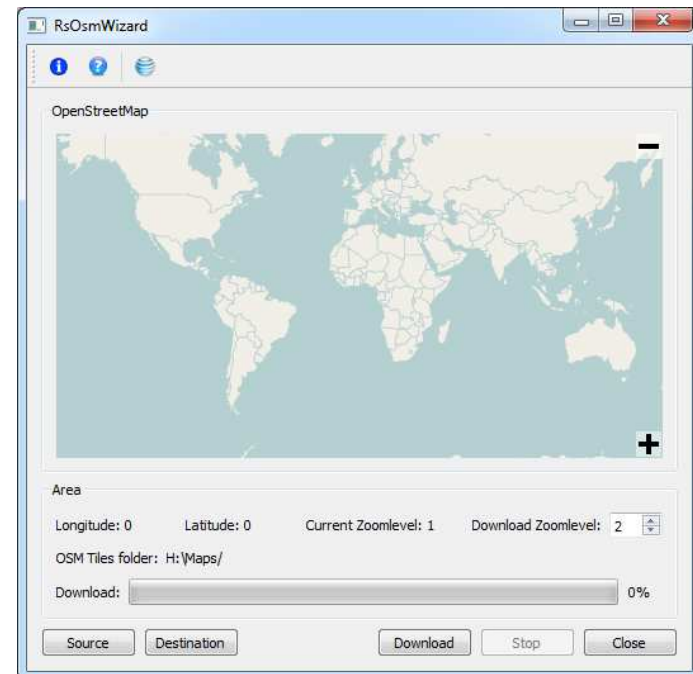
R&S PR200 Firmware and Utilities CD contains RsOsmWizard PC software that allows user to download the map data.



DOWNLOADING MAP

To download maps :

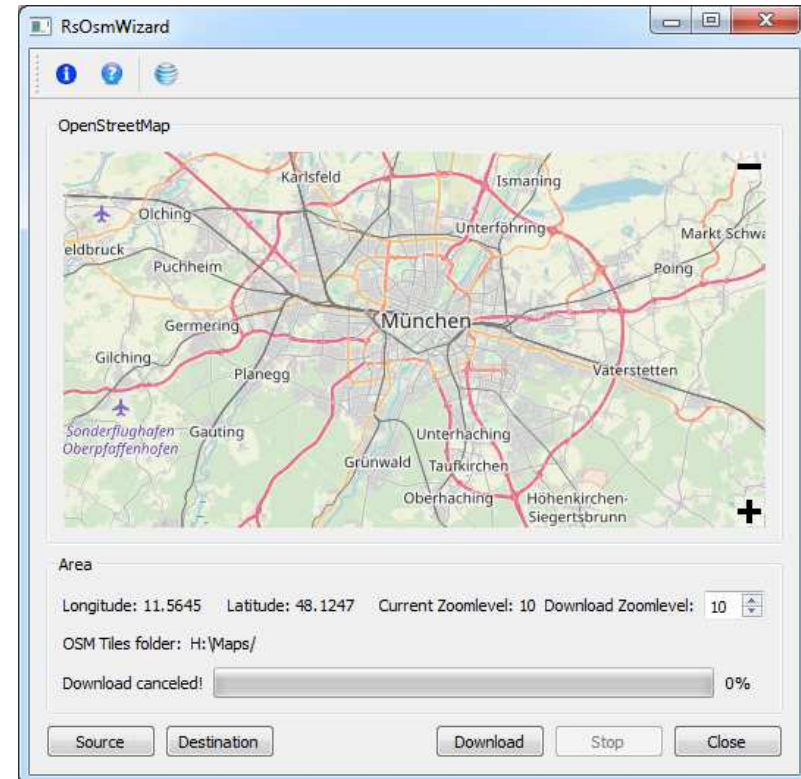
1. Install the RsOsmWizard application provided with your R&S PR200.
2. Run the RsOsmWizard application.
3. Insert a formatted SD card (FAT32 or exFAT format) into your PC's card reader.
4. Click "Destination". A file browser opens.
5. Navigate to the SD card entry.
6. Create a folder called Maps (if not already available).
7. Select the Maps folder.



DOWNLOADING MAP

To download maps :

8. Zoom in to the country level and click download.
9. Zoom in to city level and click download.
10. Repeat 9. for different zoom levels.
11. Close the software when done.
12. Check that the file map.prj is present in the map folder on the SD card.



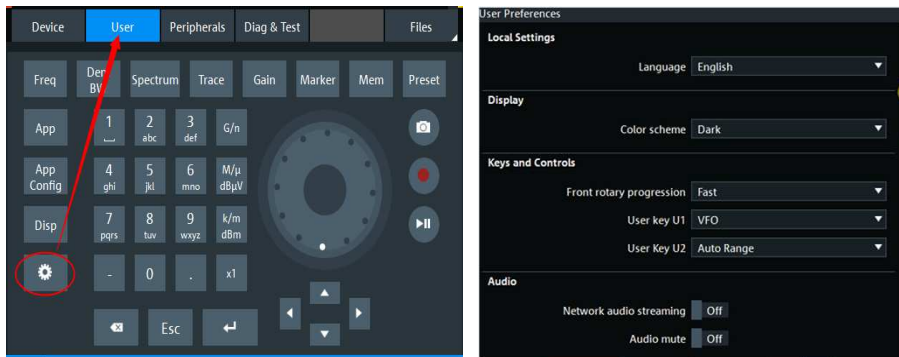
ADD MAP INTO R&S PR200

To add downloaded map into R&S PR200:

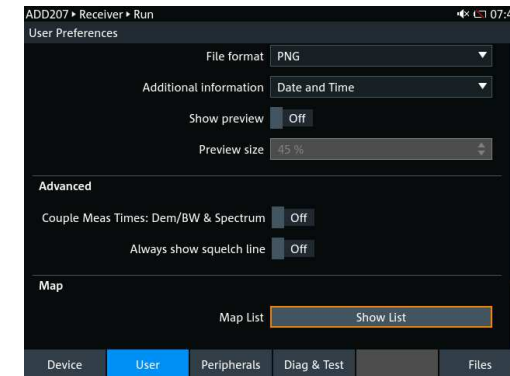
1. Insert the SD card (containing the map files) into the card-slot of the R&S PR200.



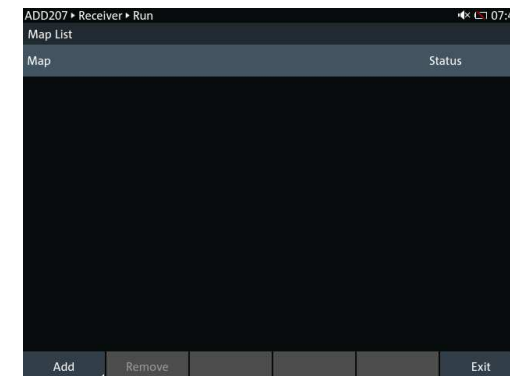
2. Press [Device Config] and then "User". The "User Preferences" page appears in the display.



1. Scroll down to the "Map" section.

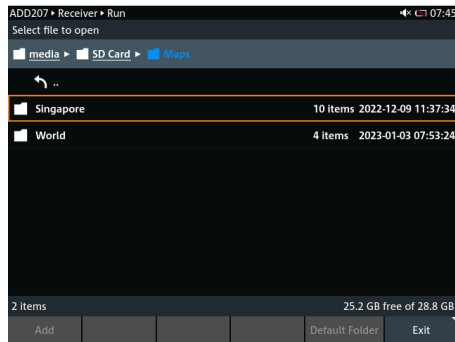


2. Select "Show List" and then press the center of the front rotary control.

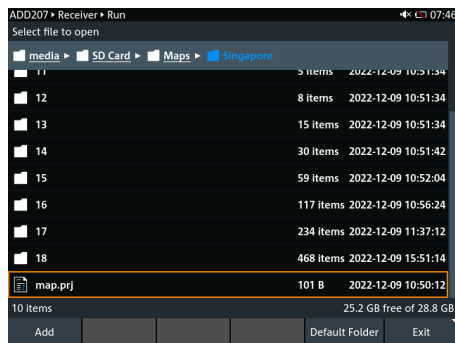


ADD MAP INTO R&S PR200

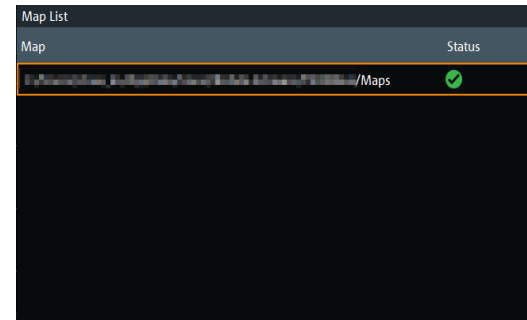
- Press "Add". The contents of the SD card's Maps folder appear as a list.



- Select the folder of the downloaded map and press the center of the rotary control.



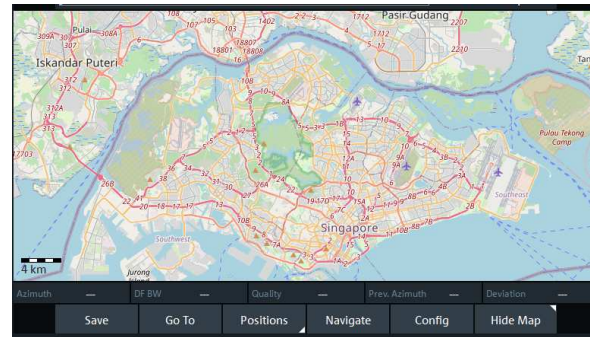
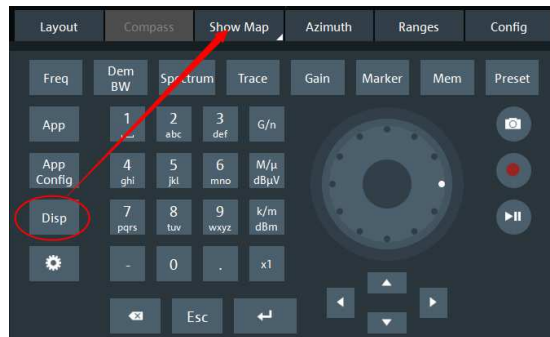
- Select "map.prj" file and press "Add". A new entry appears in the "Map List".



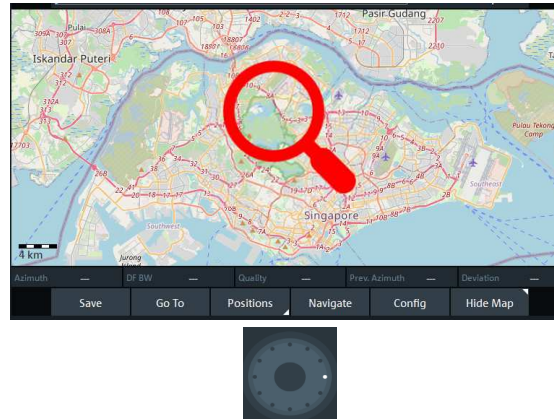
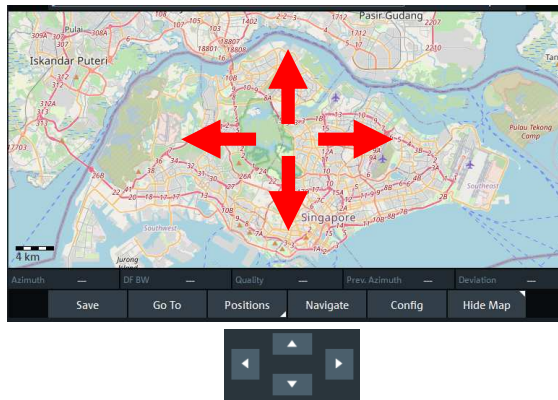
- Press "Exit".

TO SEE THE MAP IN R&S PR200

1. Press “Disp” then “Show Map”.



2. Use the arrow keys to pan map and the rotary control to zoom in and out.





HANDS ON

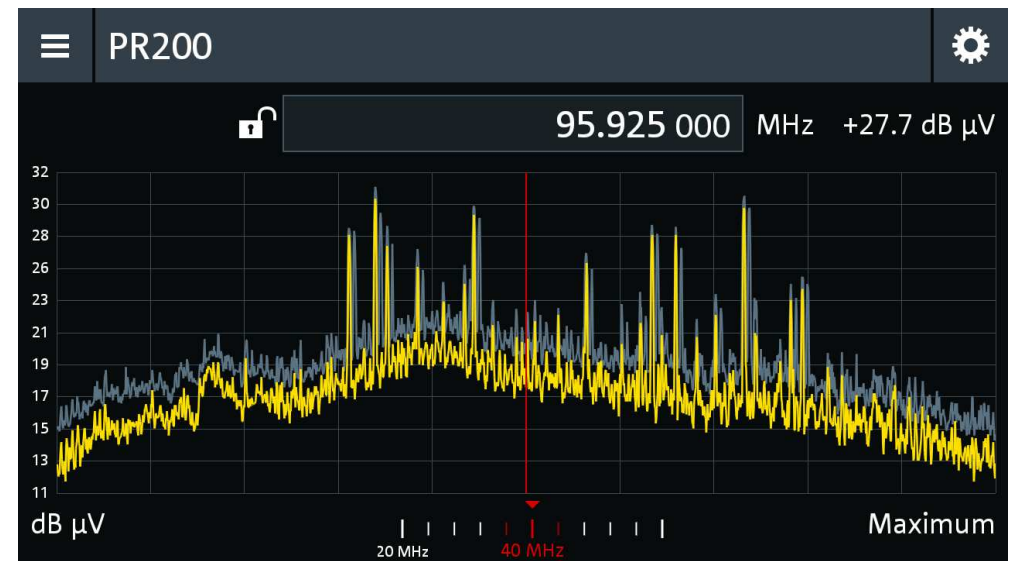
- ▶ Receiver Touch & Feel
- ▶ Antenna Touch & Feel
- ▶ PR200-HE400 Setup
- ▶ PR200 1st ON
- ▶ *(est. 15 - 25 min)*



MOBILE APP INTEGRATION WITH PR200

PR200 ANDROID APP

- ▶ The R&S PR200-App controls basic functions of your PR200.
- ▶ The app displays the spectrum as would be seen on the PR200, allowing for mobile viewing of signals away from the receiver.
- ▶ Allows you to set,
 - Frequency
 - FFT-Mode
 - Span
 - Maxhold
 - Demodulation
 - And other settings.
- ▶ You can scroll through frequencies and select peaks to center them in the spectrum by using simple gestures.



PR200 ANDROID APP - SETUP

- ▶ Connect the provided **TPLink Archer T2U Nano** Wireless USB Adapter into the USB port of the PR200.
- ▶ Power on the PR200 after



- ▶ Connect to the new Wi-Fi hotspot provided by the Archer Wireless adapter and PR200. The login information is as follows,
 - SSID: *PR200*
 - Password: *PR200PR200*

- ▶ Install the RP200 App from the Google Play store on a compatible android smartphone

R&S®PR200

Rohde & Schwarz GmbH & Co. KG

500+ Downloads Rated for 3+ ⭐

Install

Share

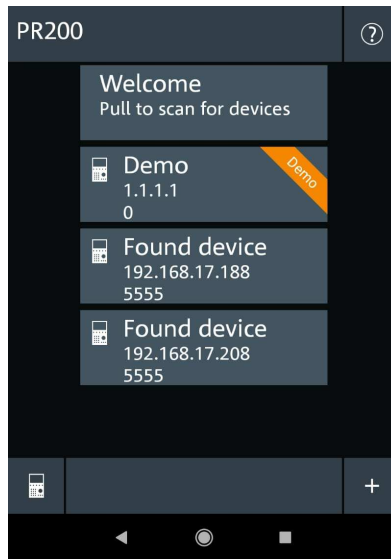
Add to wishlist



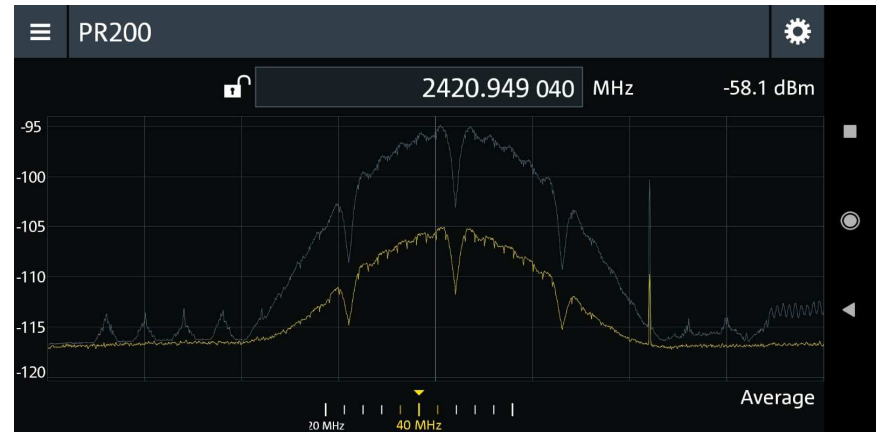
- ▶ Once Installed, run the PR200 Application on the smartphone.
- ▶ OR, if the PR200 is already connected to an existing network, we can use the same network for the app connection.

PR200 ANDROID APP - SETUP

- ▶ Within the PR200 App, Slide down to begin the automated search for PR200 devices. Once done, connect to the IP address corresponding to the PR200



- ▶ The spectrum will show up on the app once connected

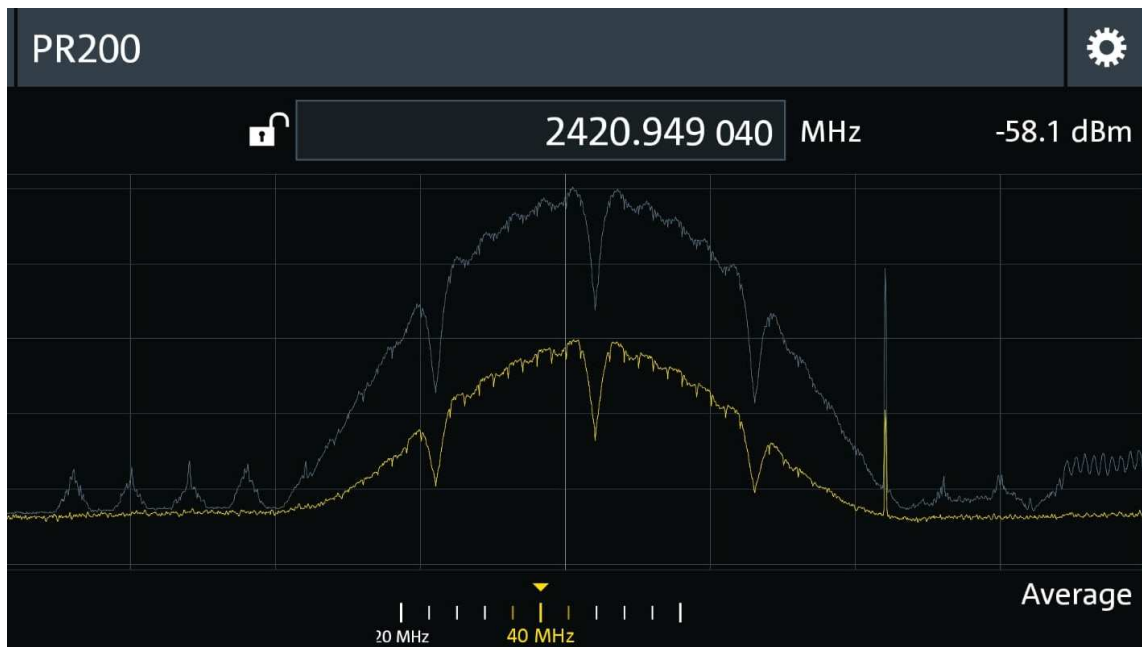


PR200 ANDROID APP - CONFIGURATION

- Control of the App can be seen within the Settings Page.

Settings	
General Settings	
Frequency:	2420.949 040 MHz
Display Max Hold:	☒
Audio:	☐
FFT Detector:	Average
Detector:	Peak
Demodulation:	FM
Demodulation Bandwidth:	40 MHz
Disconnect	IP: 192.168.17.208 Port: 5555





HANDS ON

► PR200 Setup

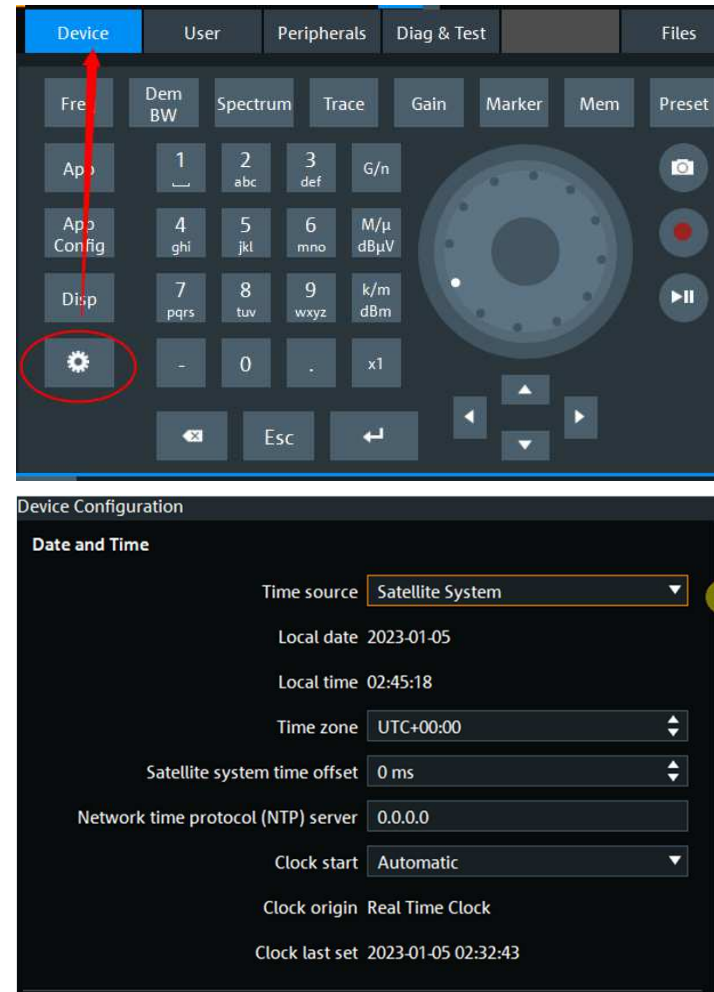


PR200 BASIC CONFIGURATION & SETTINGS

PR200 SETTINGS

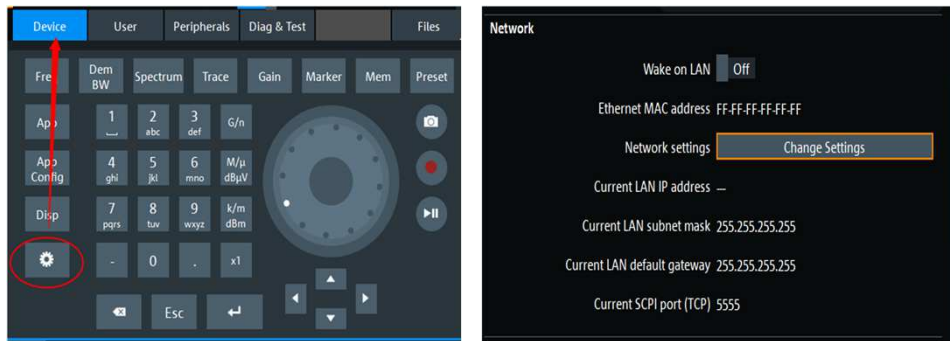
“Settings” - “Device Configuration”

- Date and time and network parameters can be configured.
- To check on the available options and versions of your PR200

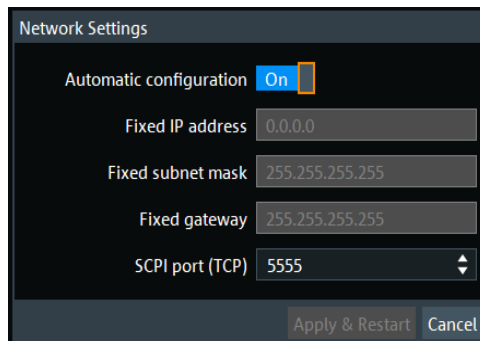


NETWORK SETTING

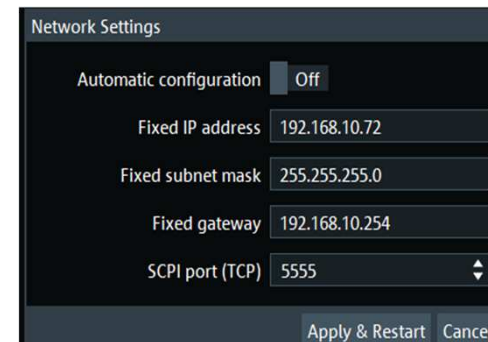
1. Press the “Device Config”, “Device” and then scroll down to “Network” section.



2. Select “Change Settings” and press the center of the rotary control.



3. Turn off “Automatic configuration” and enter the static IP address, subnet mask and gateway.

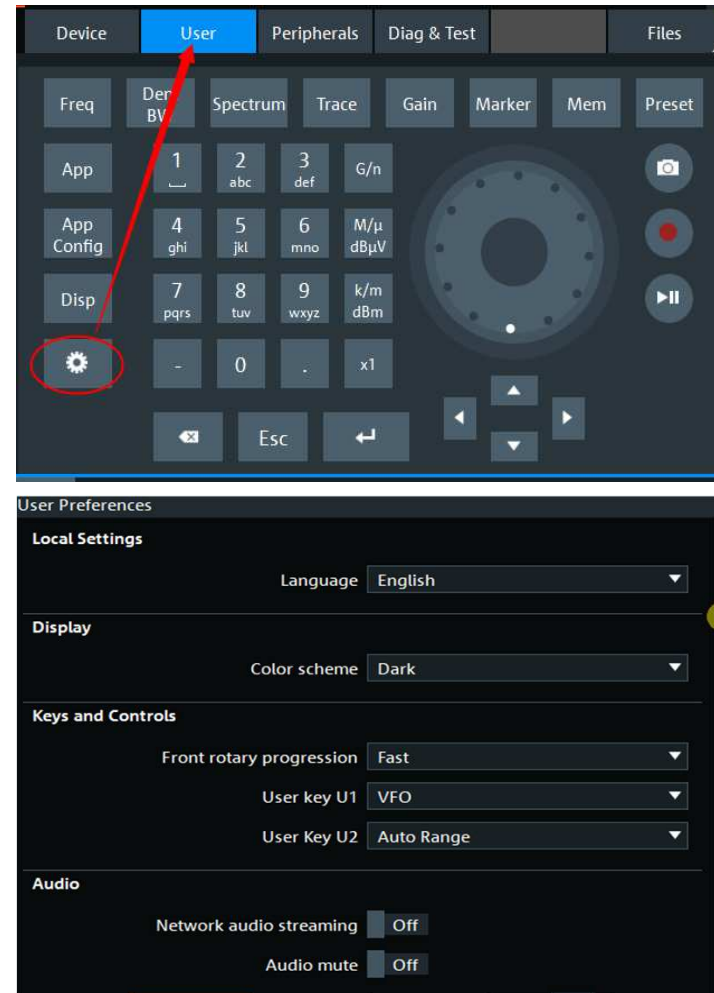


4. Press “Apply & Restart” which will restart R&S PR200 to register the new IP address.
5. **NOTE:** only when there is an active connection will the current IP be shown. The Network setting can still be changed even when the current IP is blank.

PR200 SETTINGS

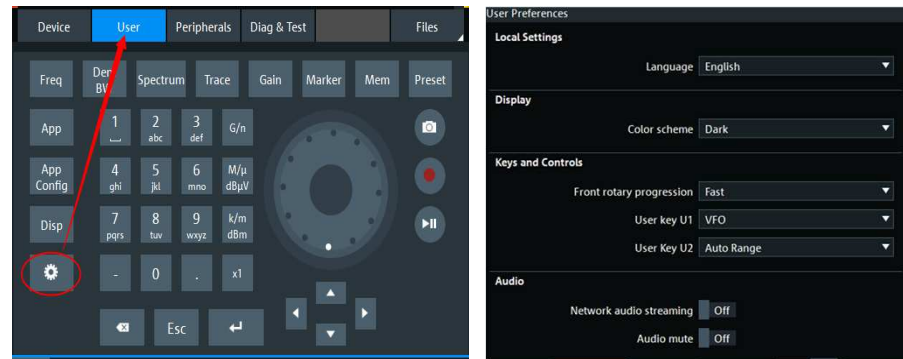
“Settings” - “User Preferences”

- Here the following can be set:
 - Language (English or 简体中文)
 - Display color scheme
 - User keys (U1/U2) assignment
 - Audio settings
 - Tone settings
 - Data and file formats
 - Location Panel toggle
 - Map list

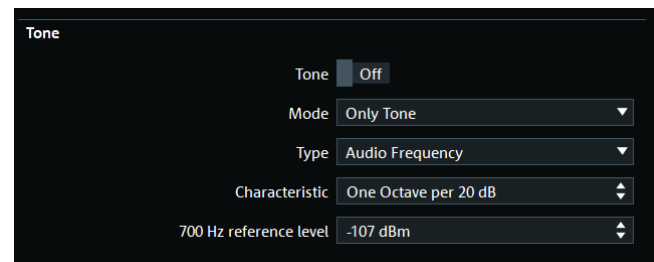


ENABLE TONE

1. Press [Device Config] and then “User”. The "User Preferences" page appears in the display.



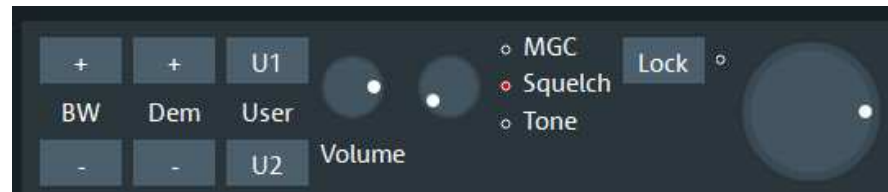
2. Scroll down to the “Tone” section.



3. Enable “Tone” and adjust the “700 Hz reference level”.

ENABLE TONE

1. Tone can also be enabled via the top control of R&S PR200.

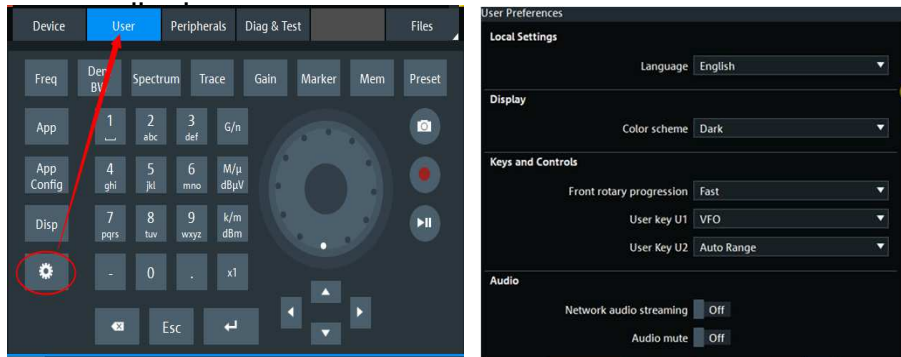


2. Press the “MGC/Squelch/Tone” knob until the corresponding “Tone” LED lights up.

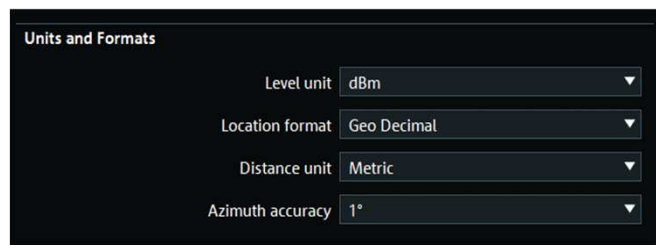
3. Turn the knob to adjust the “700 Hz reference level”.

CHANGING UNITS AND FORMATS

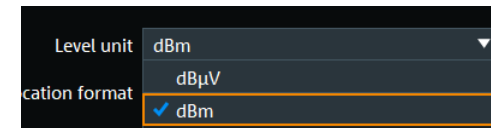
1. Press [Device Config] and then “User”. The “User Preferences” page appears in the



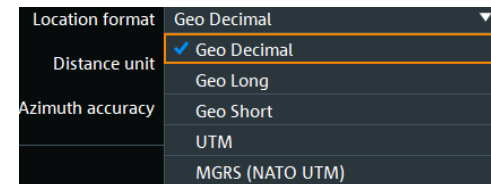
2. Scroll down to the “Units and Formats” section.



3. Level unit can be changed by changing the “Level unit” field. Available units are dBm and dBuV.

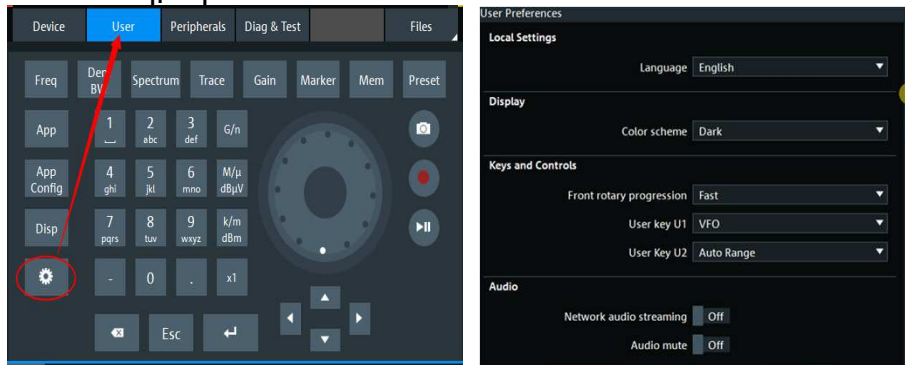


4. Co-ordinates format can be changed via “Location format” field.

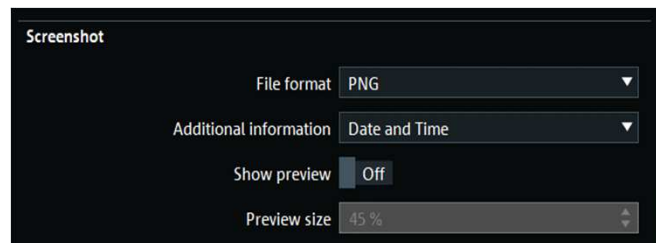


SCREENSHOT FORMAT

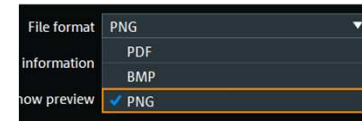
1. Press [Device Config] and then "User". The "User Preferences" page appears in the



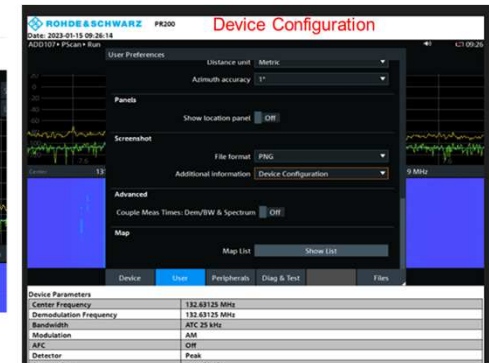
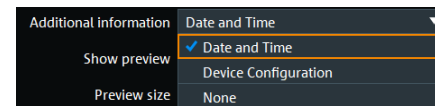
2. Scroll down to the "Screenshot" section.



3. Level unit can be changed by changing the "Level unit" field. Available units are dBm and dBuV.

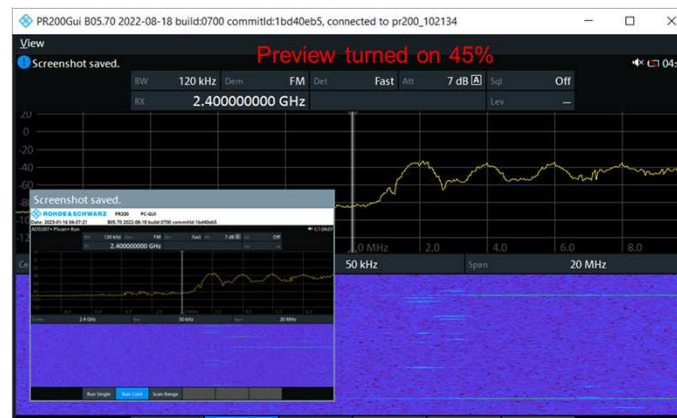
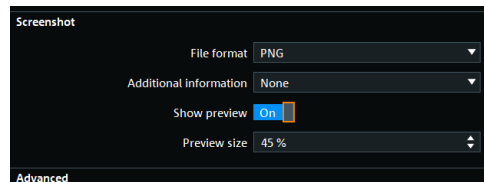


4. Additional information can be included with the screenshot.



SCREENSHOT FORMAT

5. To show preview of the screenshot.

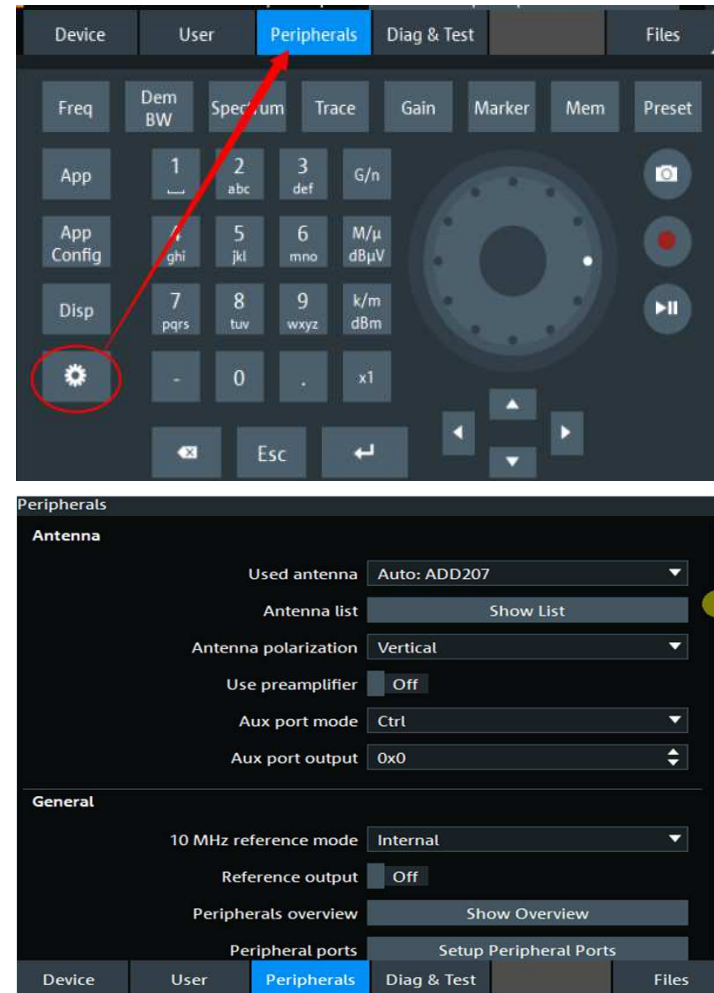


PR200 SETTINGS

“Settings” - “Peripherals”

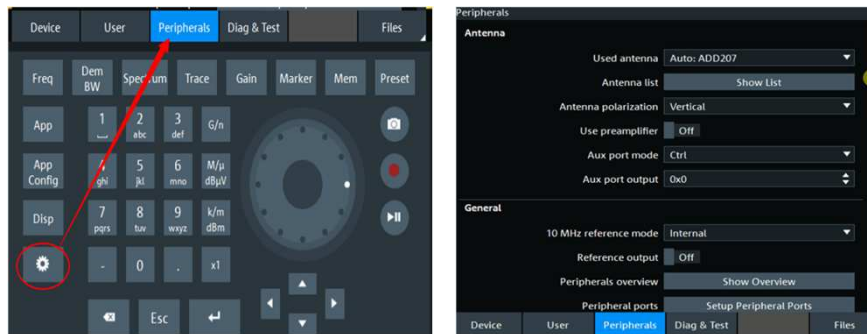
► Here the following can be set:

- Antenna configuration
- GPS and compass sources
- Coordinate formats
 - Geo Decimal
 - Geo Long
 - Geo Short
- UTM
- MGRS (NATO UTM)

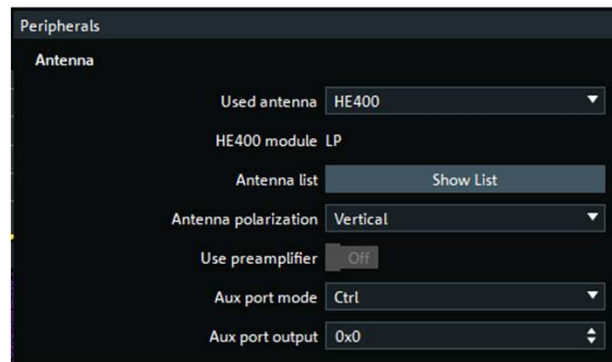


TO CHECK THE CURRENTLY USED ANTENNA

1. Press [Device Config] and then “Peripherals”. The “Peripherals” page appears in the display.

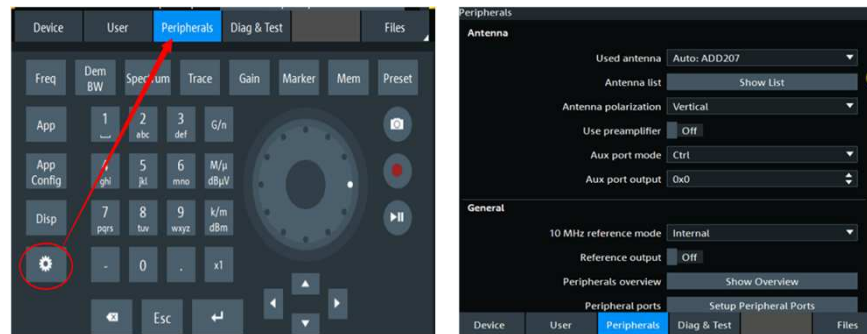


2. Under “Antenna”, ensure that the correct antenna is reflected here. If not selected, select the antenna, HE400 or HE800, under the Used Antenna dropdown box.



TO CHECK ANTENNA SETTINGS WITHIN THE PR200

1. Press [Device Config] and then “Peripherals”. The “Peripherals” page appears in the display. Select “show List” under antenna list



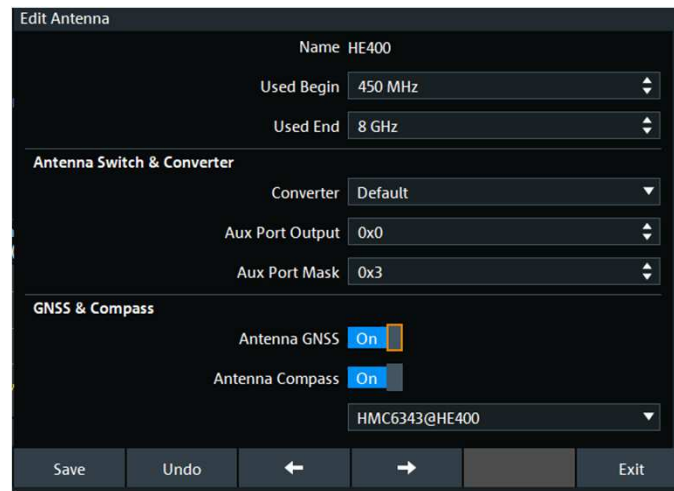
2. In the new window, scroll to the antenna being used and select ‘edit’.

The screenshot shows the 'Antenna List' window with a table containing one entry. At the bottom, there are buttons for 'Add', 'Edit', 'Remove', 'Refresh', and 'Exit'.

Name	Ant. Type	Used Begin	Used End
HE400	RX	450 MHz	8 GHz

TO CHECK ANTENNA SETTINGS WITHIN THE PR200

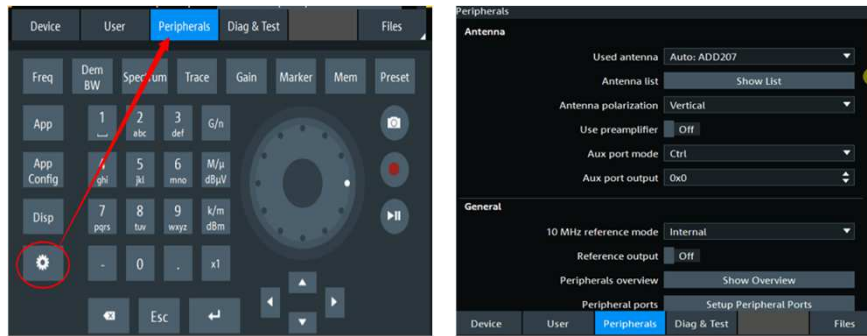
3. Check that all information within this page corresponds to the antenna used. Pay attention to GNSS and Compass, ensure that they are on as available.



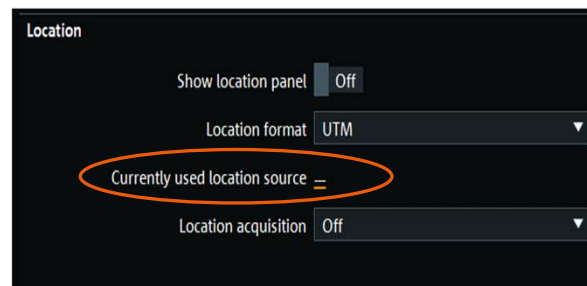
4. Once done, select “save” and “exit”

TO SEE CURRENT USED LOCATION SOURCE

1. Press [Device Config] and then “Peripherals”. The “Peripherals” page appears in the display.

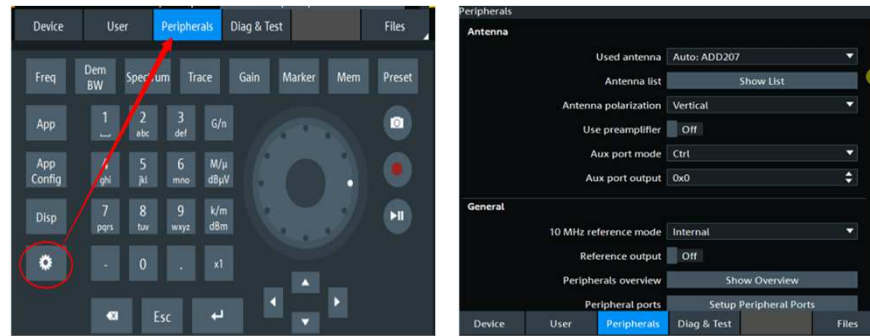


1. In the “Location” section, observe “Currently used location source”.

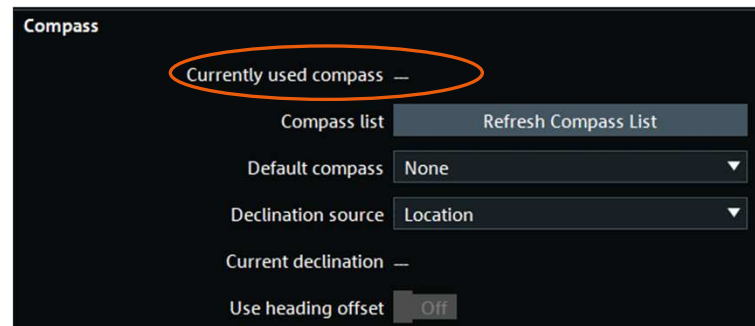


TO SEE CURRENT USED COMPASS

1. Press [Device Config] and then “Peripherals”. The “Peripherals” page appears in the display.



1. In the “Compass” section, observe “Currently used compass”.

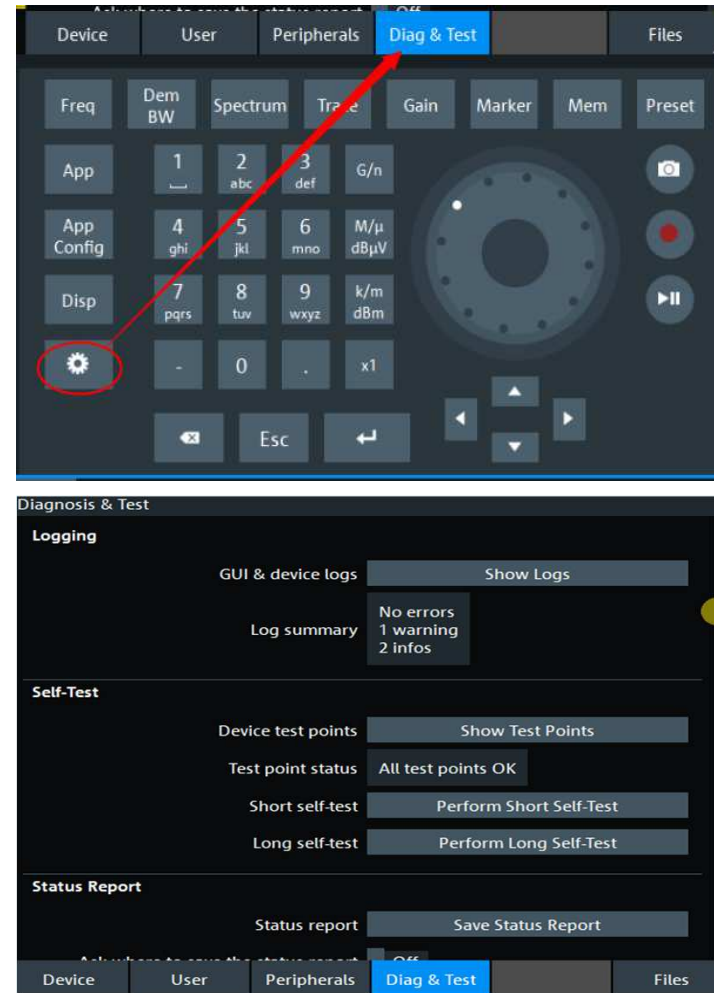


PR200 SETTINGS

“Settings” - “Diagnosis & Test”

► Here contains the following:

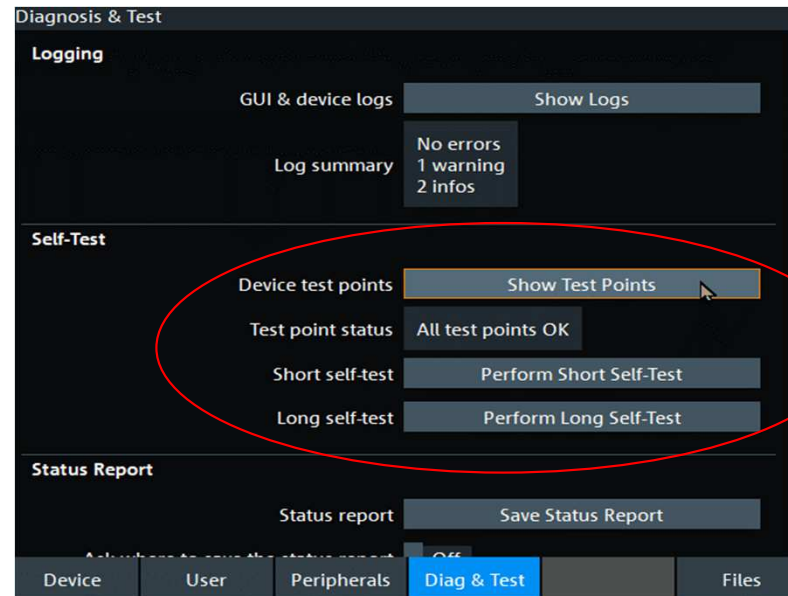
- Logs.
- Activate Short or Long self-tests.
- Device status report.
- Device continuous test points status.



PR200 SETTINGS

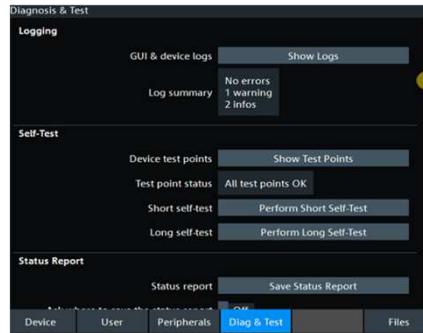
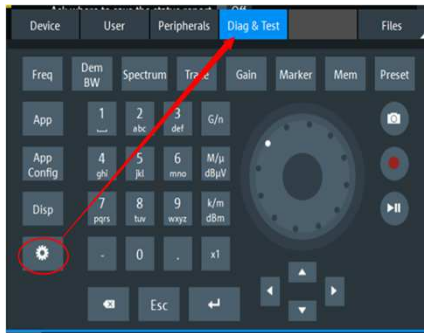
“Settings” - “Diagnosis & Test”

- ▶ Check that “Test point status” is “OK”.
- ▶ Run the “Short Self-Test” or “Long Self-Test” to check the internal RF paths.
- ▶ “Short” for single frequency.
- ▶ “Long” for comb frequencies.

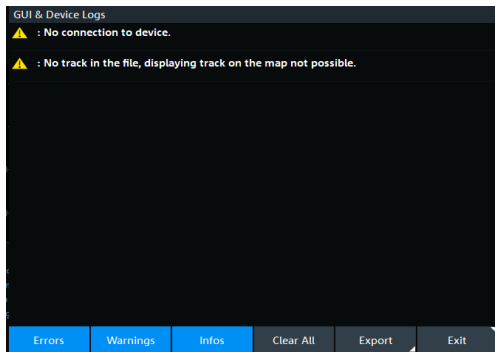


EXPORT LOG

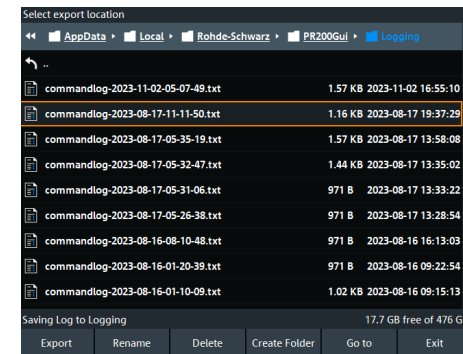
1. Press [Device Config] and then “Diag & Test”. The “Diagnosis & Test” page appears in the display.



2. In the “Logging” section, select “Show Logs” and press the center of rotary control. GUI & Device Logs window opens.



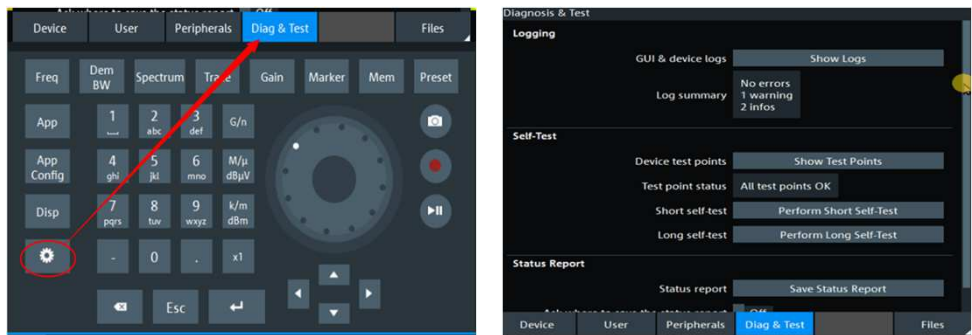
3. Press “Export”. “Select export location” dialog opens.



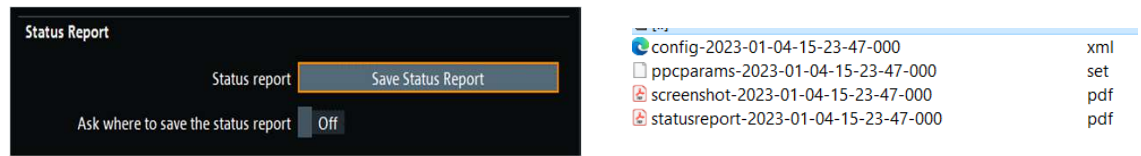
4. Choose the location and press “Export”. Default location is “Logging”.
5. Press “Exit” twice to exit “Show Logs” dialog.

SAVE STATUS REPORT

1. Press [Device Config] and then “Diag & Test”. The “Diagnosis & Test” page appears in the display.



2. In the “Status Report” section, select “Save Status Report” and press the center of rotary control. Report will be saved to SD card drive “Status Reports” folder.

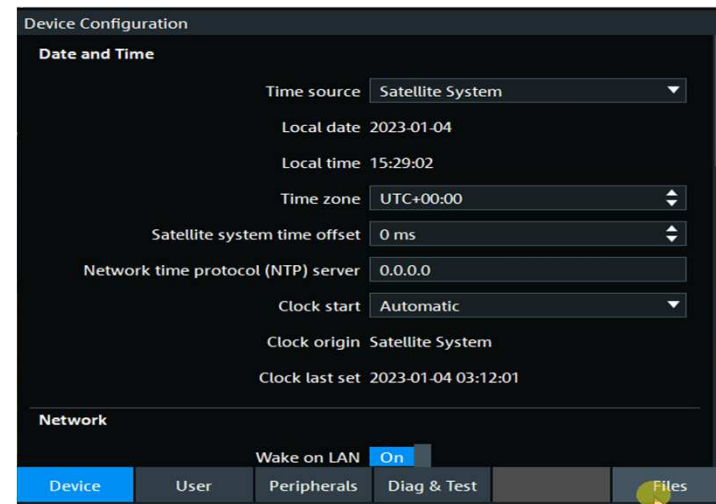
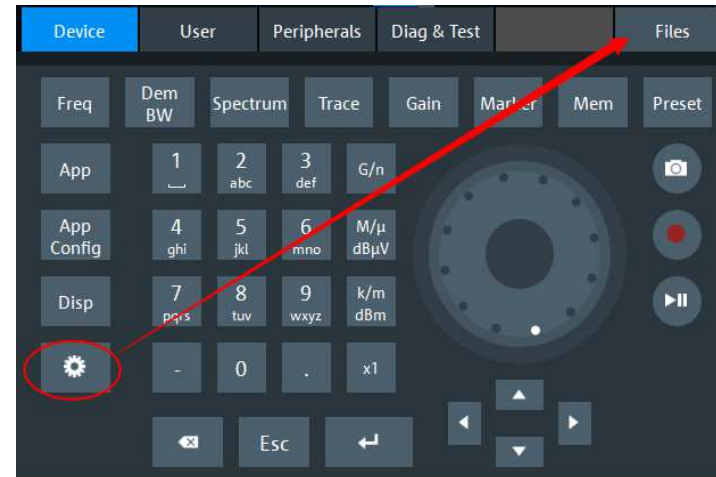


3. Choose the location and press “Export”. Default location is in SD card drive “Logging” folder.
4. Press “Exit”

PR200 SETTINGS

“Settings” - “Files”

► Allows user to manage files and folders.





HANDS ON

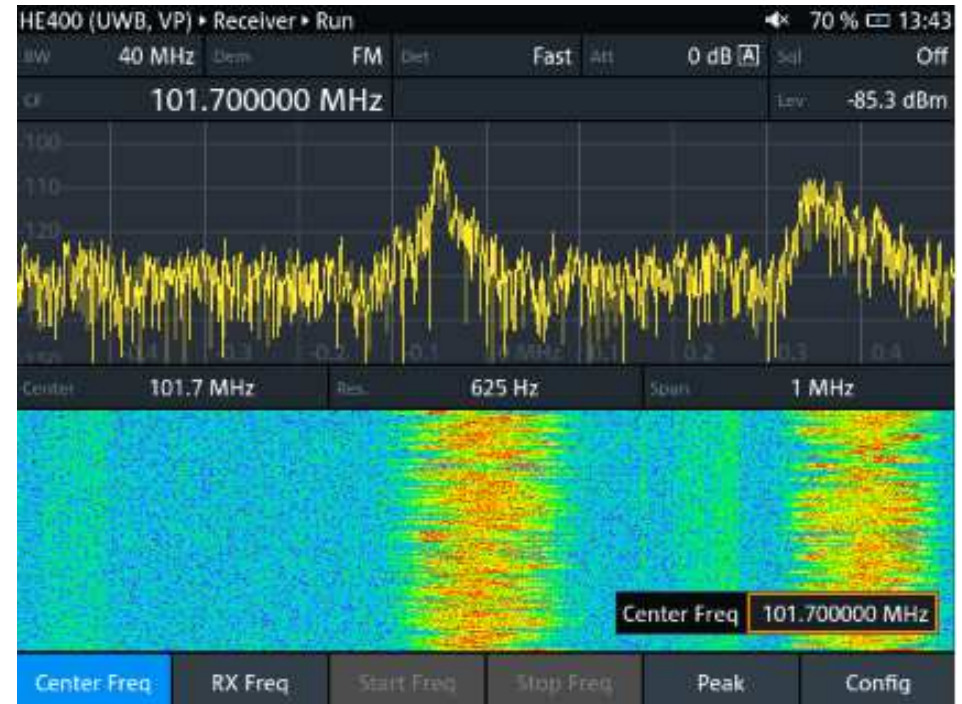
- ▶ PR200 Settings Check
- ▶ GPS & Antenna Setup
- ▶ PR200 Self-test
- ▶ PR200 files demo
- ▶ *(est. 20 - 30 min)*

RECEIVER MODE

- ▶ Gives the user the ability to examine up to 40 MHz spectrum in Real-Time
- ▶ 100% Probability of Intercept (PoI) of signals $>0.75 \mu\text{s}$
- ▶ Best choice for gap-free detailed analysis if signal/spectrum of interest is less than 40 MHz wide

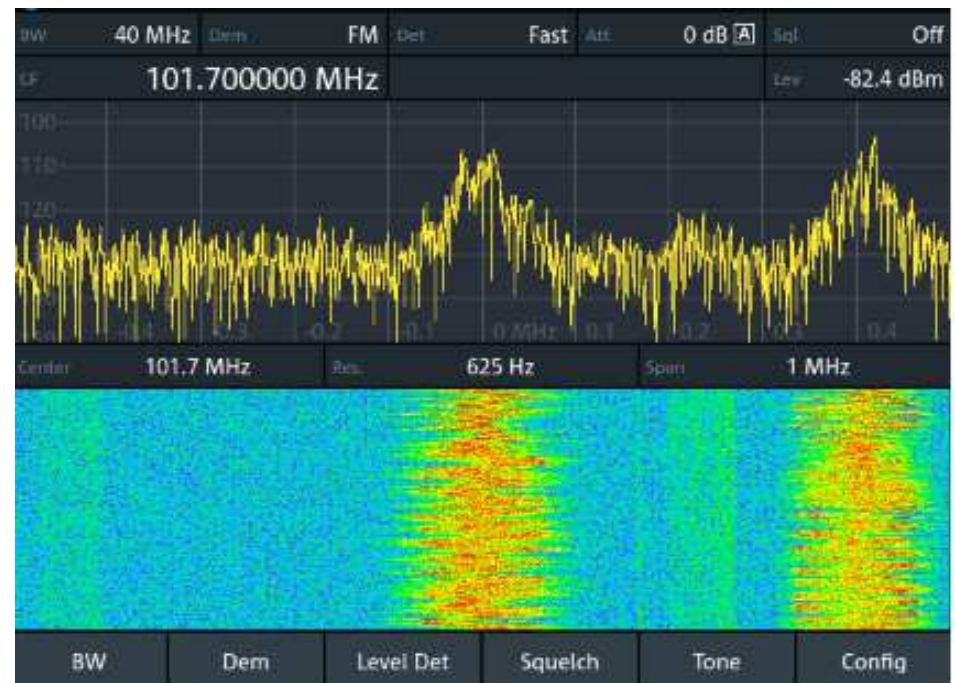
FREQUENCY

- ▶ Center Freq - Sets the center frequency of the PR200
- ▶ RX Freq - Sets the demodulation frequency for, must be within current view (101.2 - 102.2 MHz in this picture, 101.7 MHz +/- 500 kHz)
- ▶ Peak - Quickly move across the display with arrow keys
- ▶ Config - Turn Automatic Frequency Control (AFC) on/off
 - Useful for signals that drift



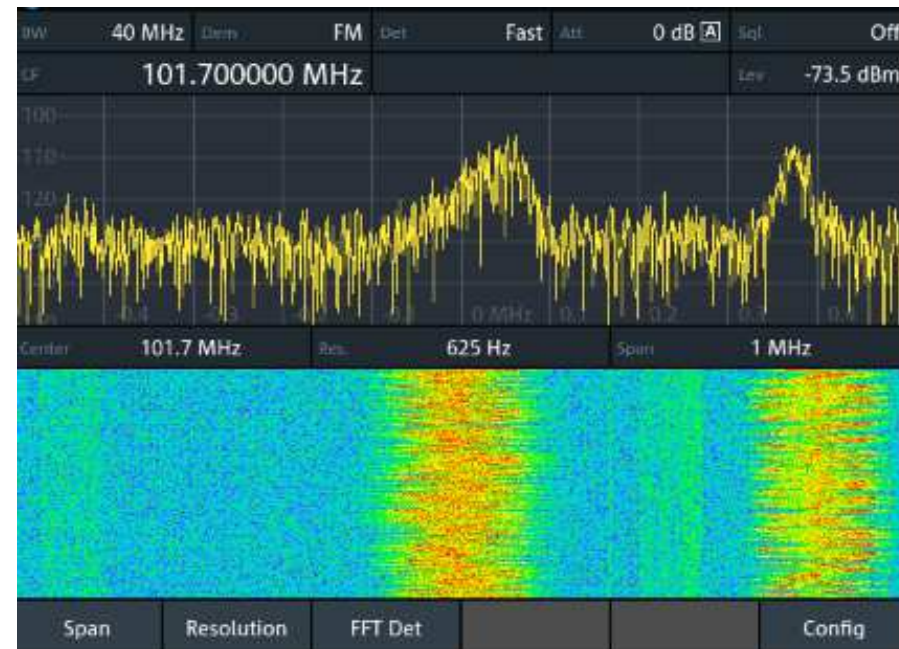
DEMODULATE

- ▶ BW - Set Demodulation BW
- ▶ Dem - Set Demodulation Type
- ▶ Level Det - Set Detector Type
 - Peak - Highest Value
 - RMS - Root Mean Square
 - Average - Average
 - Fast - Last Value Collected
- ▶ Squelch - Enable/Disable and set the Squelch Level
- ▶ Tone - Turn Tone Mode On/Off
- ▶ Config - Set Measurement Time, Auto is always fine



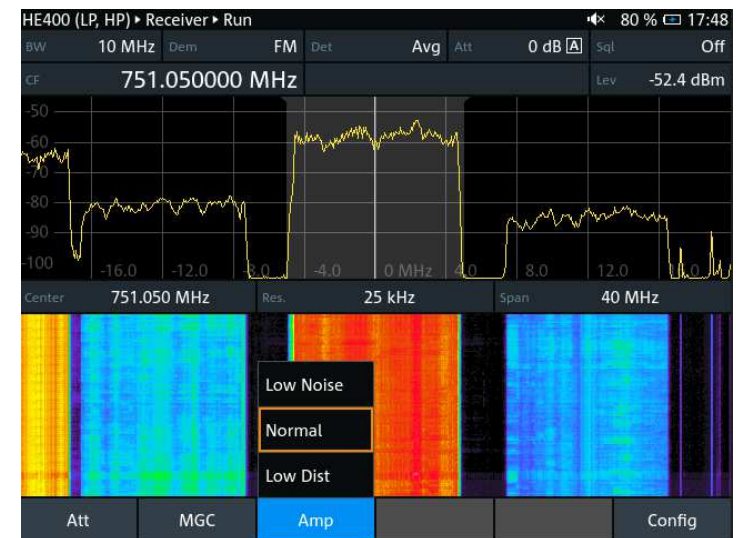
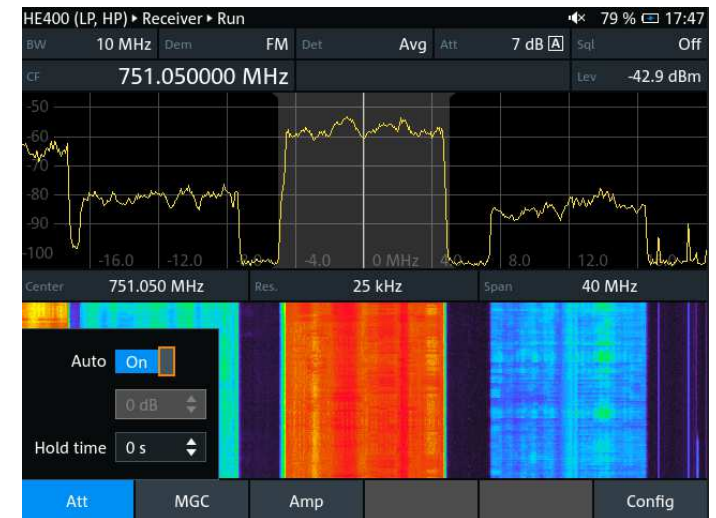
SPECTRUM

- ▶ Span - Sets the Span from center frequency that is shown, up to 40 MHz
 - i.e. 20 MHz span gives you +/- 10 MHz from center
- ▶ Resolution - Sets Frequency Resolution, leave on auto
- ▶ FFT Det - Enable/Disable traces or change types. The PR200 can have 1-3 simultaneous traces active:
 - Auto Peak: Draws line between High and Low Spectrum Values over Meas Time
 - Positive Peak: Max Spectrum over Meas Time
 - Negative Peak: Min Spectrum over Meas Time
 - Average: Average Spectrum over Meas Time
 - RMS: RMS Spectrum over Meas Time
 - Sample: Instantaneous Spectrum (Clear Write)
 - Off: Disables Trace (Only for traces 2 and 3)



ATTENUATOR SETTINGS

- ▶ The PR200 contains tools to help it prevent strong signals in the environment from creating intermodulation products inside the device
 - Built In Attenuator (0 to 40 dB in 1 dB steps) with Manual or Auto Mode
 - Receiver Mode:
 - Low Noise: 2 Preamplifiers Active (Most Sensitive)
 - Normal: 1 Preamplifier Active
 - Low Distortion: 0 Preamplifiers Active (Most Resistant to Strong Signals)



THANK YOU

- ▶ Moving onto the next section:
- ▶ PR200 APPs

